

Probability, Statistics & Data Analysis

Lecture 7: t-tests and assumptions

**All models are wrong,
but some are useful**

George Box

Lecture 7: t-tests and assumptions

What we will cover

- Degrees of freedom
- Confidence intervals with t-distribution
- T-tests
 - One sample t-test
 - Independent two-sample t-test
 - Dependent two-sample t-test
- Assumptions
 - Normality
 - Homoscedasticity
- Transformation
- Lognormal distribution
- Outliers





When we don't know population standard deviation

Basically always!

Accounting for variability in sample statistics

- In Lecture 6, we estimated how likely it was that we observed what we did if the null hypothesis was true
- We used statistical methods to account for uncertainty that arose through use of

the sample mean, $\bar{x}$, as an estimate of the population mean, μ

- But, we assumed we knew the population standard deviation, such that the standard error on the mean was $\sigma/\sqrt{n}$
- In practice, this is rarely ever the case
- We therefore also need to be able to account for the uncertainty associated with

use of the sample standard error on the mean, $s/\sqrt{n}$, as an estimate of the population standard error on the mean, $\sigma/\sqrt{n}$

$$\sigma/\sqrt{n} \approx s/\sqrt{n}?$$

When is it reasonable to assume $\sigma_{\bar{X}} \approx s_{\bar{X}}$?

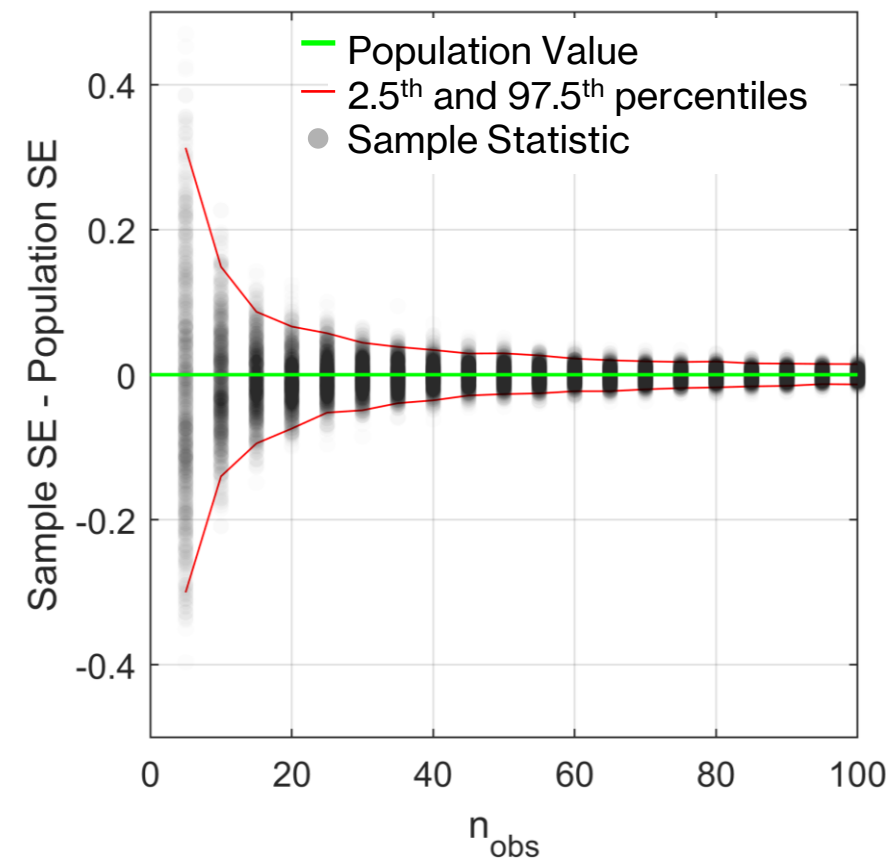
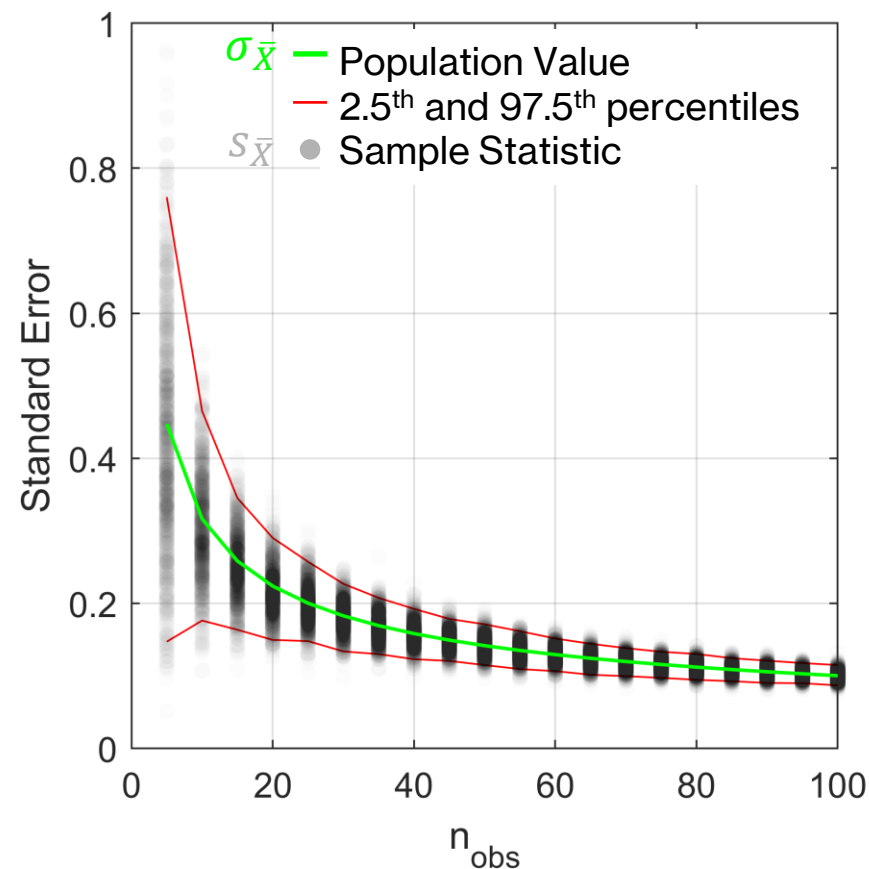
- The only option we have is to use the sample SEM ($s_{\bar{X}}$) as an estimator of the population SEM ($\sigma_{\bar{X}}$)
- We know intuitively that as sample size increases, the precision of the estimate will increase

How large a sample do we need for the *sample* SEM to be a sufficiently accurate estimate of the *population* SEM?

- Or mathematically, for $\sigma_{\bar{X}} \approx s_{\bar{X}}$
- The literature would have you believe $n = 30$ or 40 is an appropriate threshold
- I ran some simulations to investigate
 - $\sigma = 1$
 - $B = 1000$
 - $n_{\text{obs}} = 5:5:100$ (5, 10, 15 ,..., 95, 100)

The effect sample size (n_{obs}) on $\sigma_{\bar{X}} \approx s_{\bar{X}}$

- Range of sample SE values decreases with sample size
- Hence precision increases as sample size increases
- No obvious threshold
- So just get as many observations as possible?



Why not just do more experiments?

Example 1: iPerfusion

- Measuring conductance of eye to investigate glaucoma pathophysiology and treatments
- Experiments need real mouse eyes
- Moral imperative to use small n

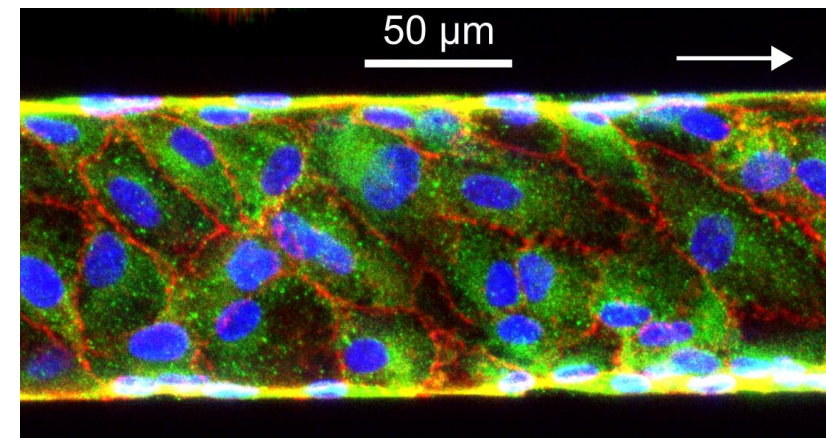
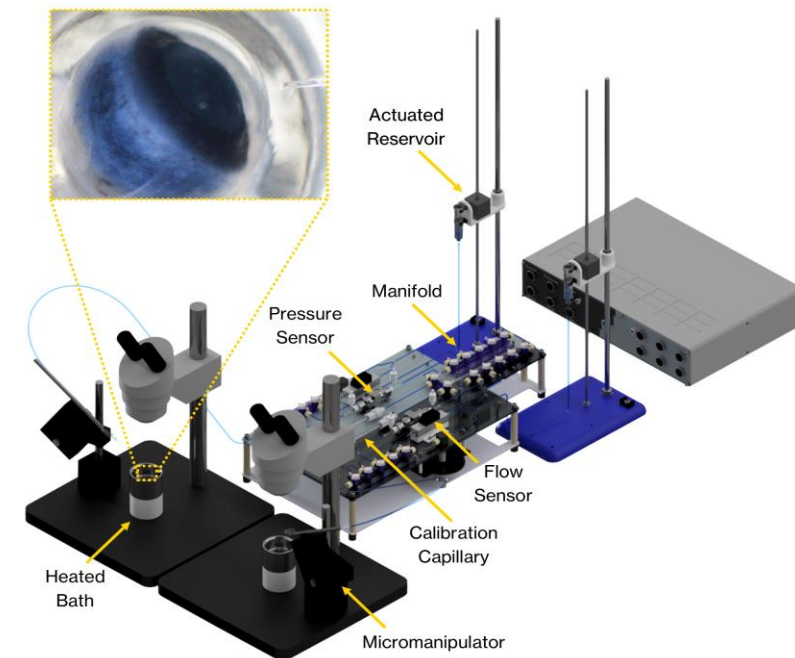
Example 2: in vitro blood vessel

Getting to a stable 'blood vessel' takes >7 days

- Equipment to perfuse vessel costs £1000s per channel
- Practicality requires small n

Solution: we turn to the t -distribution

But first, we need to understand DOF



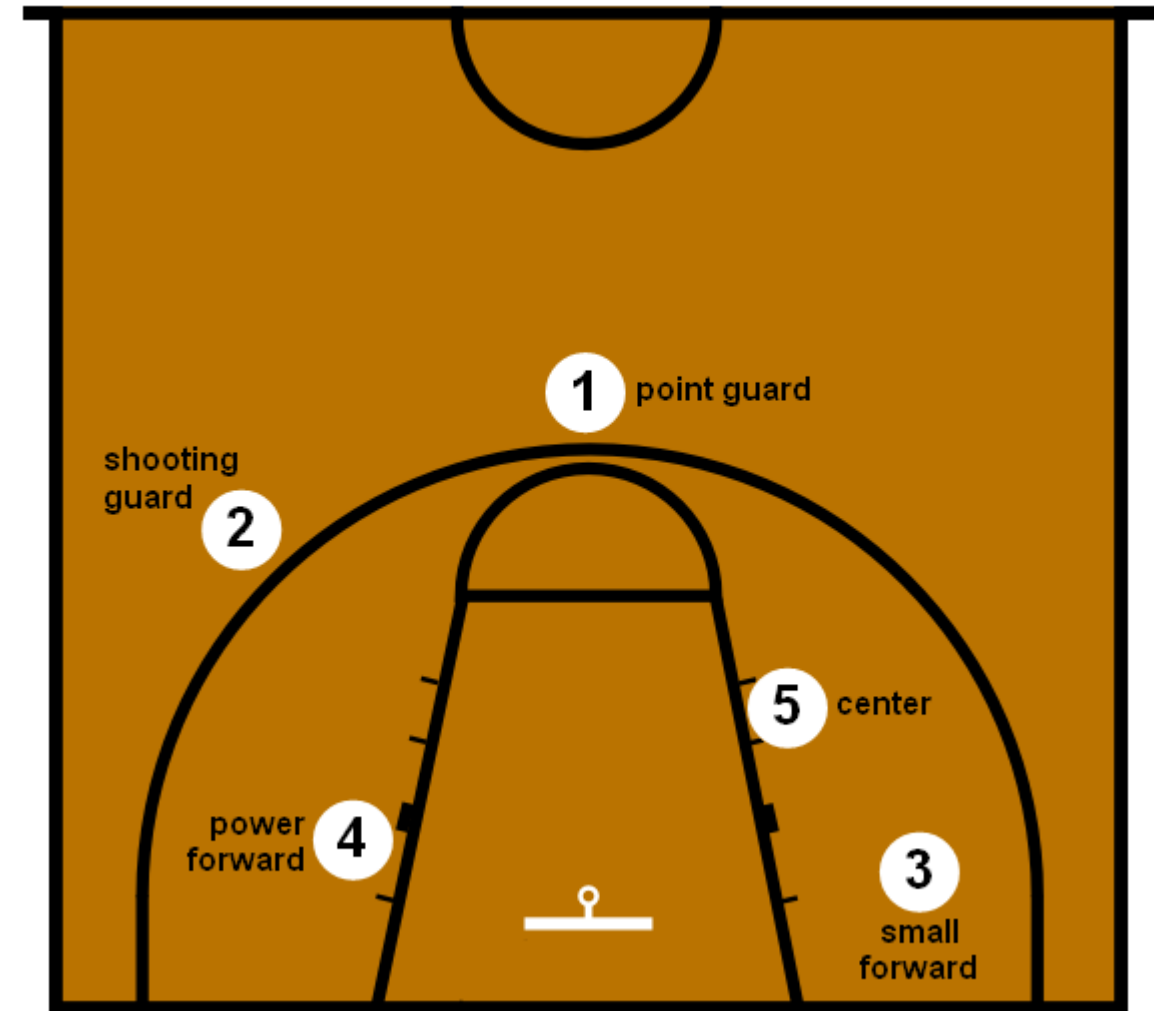


Degrees of Freedom

DOF or ν

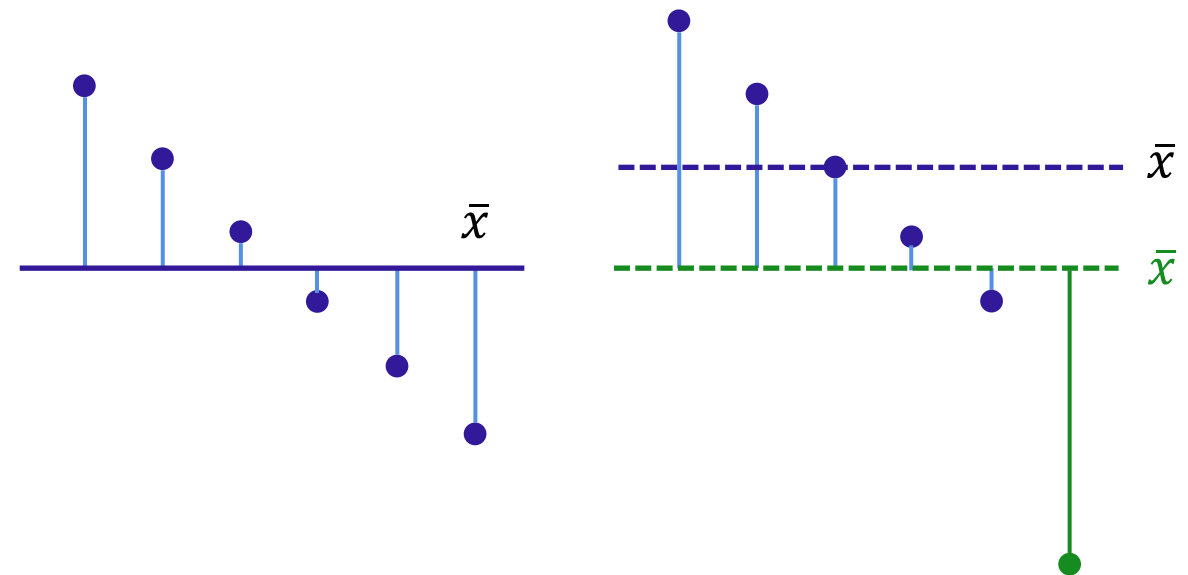
Degrees of Freedom (DOF) – Sport Analogy

- There are five players on a basketball team
- You have $N=5$ players to distribute into these positions, A, B, C, D and E
- For A, there are 5 positions you can choose
- For B, C, D there are 4, 3, 2 positions you can choose
- For E, there is no choice left
- You made $N-1$ choices
- You had $N-1$ DOF



Degrees of Freedom – Variance

- Take six values, randomly sampled from a population, e.g.
 - 0, 2, 4, 6, 8, 10
- Each value can be different (as randomly sampled)
- When calculating the mean,
 - any value can change
 - we have N DOF
- When calculating the variance
 - for a given value of the mean
 - five values can change, but the sixth must change to maintain the same mean
 - we could make N-1 changes to the data points
 - we have N-1 DOF

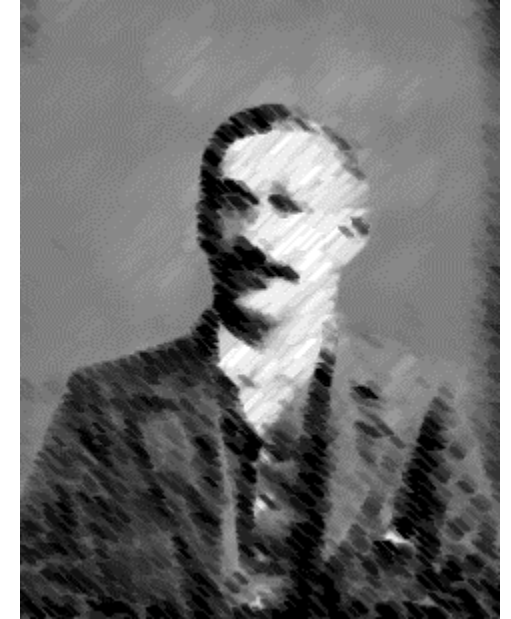




Student's t distribution

or so it's called

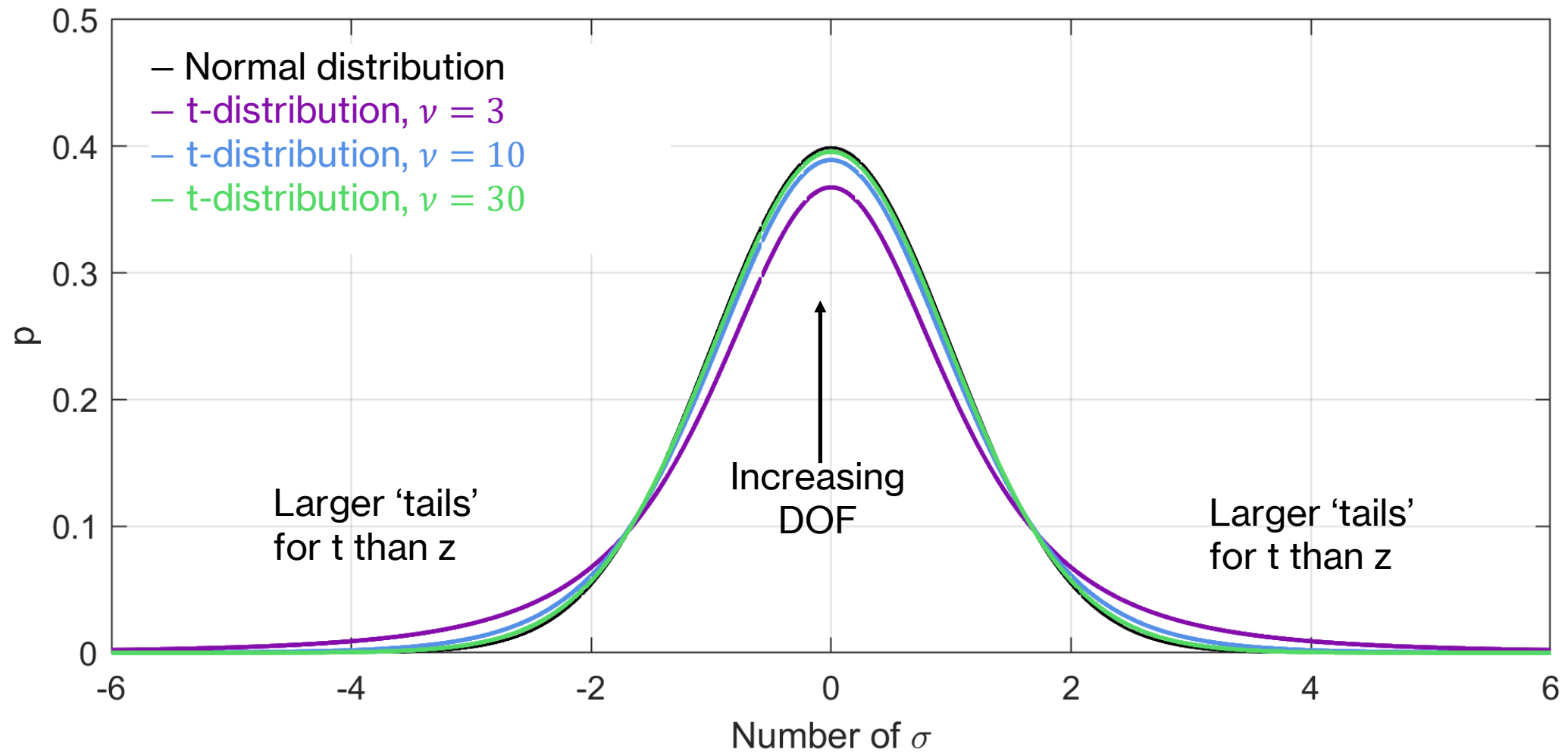
The Gosset Distribution?



- William Sealy Gosset worked at Guinness and developed the t -test and the t -statistic
- Guinness would not allow its scientists to mention Guinness, beer, or their surname
- Published under the pseudonym 'Student'
- The t -distribution is similar to the normal Z distribution,
 - It has larger 'tails' – greater probability of values away from the mean
 - It tends to the normal distribution as DOF increases

$$f(x) = \frac{\Gamma\left(\nu + \frac{1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

t vs Normal PDF



Fewer DOF – less information – more uncertainty

t-distribution CDF

- Normal distribution
- t-distribution, $\nu = 5$

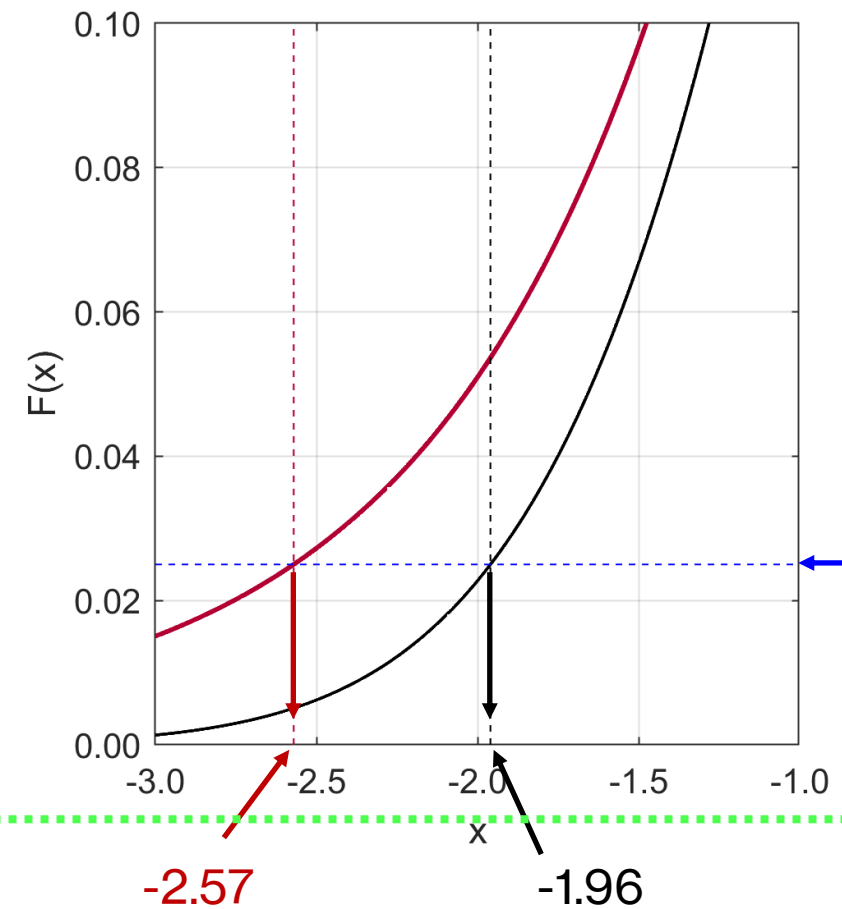
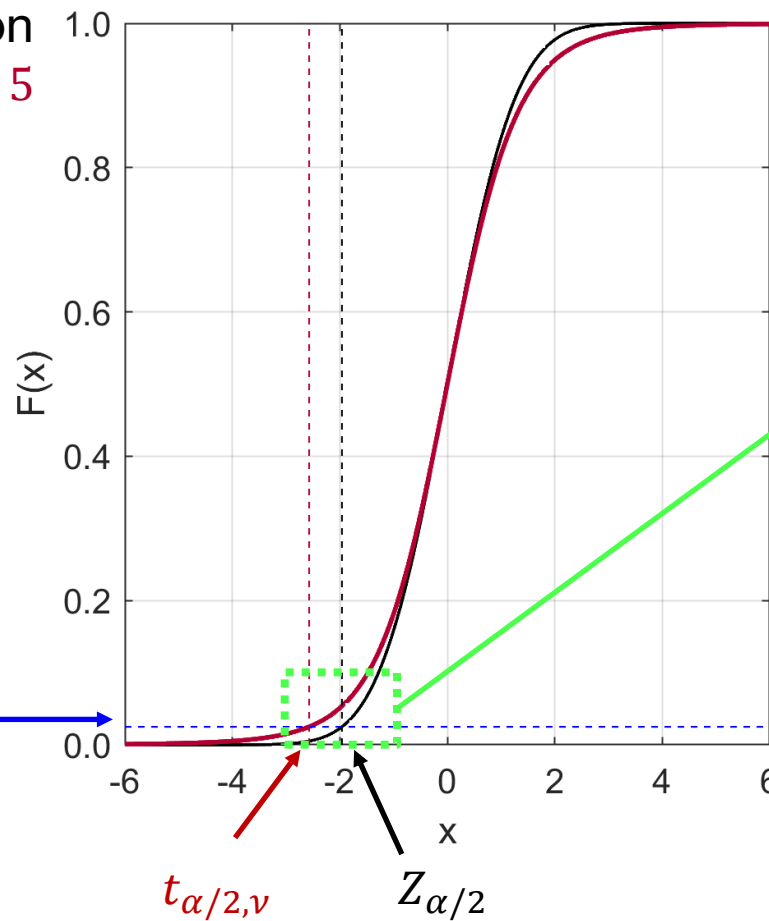
```
K>> v=5
tcrit=tinv(0.025,v)
zcrit=norminv(0.025)

v =
    5

tcrit =
   -2.5706

zcrit =
   -1.9600
```

$F(x) = 0.025$, for 95% confidence



$F(x) = 0.025$

- For a given probability, the critical t value is larger than critical z value
- For a given α , $|t_{crit}| > |z_{crit}|$ - larger test statistic is required to reject H_0

t-statistic

- Consider a random sample of n observations from a normal distribution with unknown μ and σ
- We can define the T-statistic

$$T = \frac{\bar{X} - \mu_0}{S_{\bar{X}}}$$

Has a *t*-distribution with ν degrees of freedom

$$\text{where } S_{\bar{X}} = \frac{s}{\sqrt{n}}$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

- For a single sample,

$$\nu = n - 1$$

- $n - 1$ DOF comes from the use of the sample variance to get $S_{\bar{X}}$
- With small tweaks to account for ν , we now just apply everything we know from **z-test** for the **t-test** (in blue on reference sheet)

Recap: why do we need t-distribution?

- Problem:

- z-statistic allows us to account for the fact that we use the sample mean, $\bar{x}$, as an estimate of the population mean, μ

$$Z = \frac{\bar{X} - \mu_0}{\sigma_{\bar{X}}}$$

- The z-statistic follows a normal distribution
- We also need to be able to account for the uncertainty associated with use of $S_{\bar{X}}$

- Solution

- The t -statistic

$$T = \frac{\bar{X} - \mu_0}{S_{\bar{X}}}$$

- The t -statistic follows a t -distribution with ν degrees of freedom
- It accounts for the uncertainty associated with assuming $s_{\bar{X}} \approx \sigma_{\bar{X}}$
- Note – as ν tends to infinity, $t \rightarrow z$
- **In practice, we would probably always just use a t-test/confidence interval based on t**



1 sample confidence intervals

Using the t-distribution

Intervals using the t-statistic and the t-table

- Rather than $z_{1-\alpha/2}$ (z_{crit}) we use $t_{1-\alpha/2, \nu}$ (t_{crit})
- Critical t -values can be used for confidence intervals
- MATLAB: `tinvs(1-a/2, v)`
- Or use table (noting CI is equivalent to 2-tailed)
- E.g. 90% CI with 5 samples
 - $\alpha = 0.1, \frac{\alpha}{2} = 0.05$
 - $\nu = 4$
- $t_{crit} = t_{0.95, 4} = 2.132$

$$ME_{CI} = t_{1-\frac{\alpha}{2}, \nu} s_{\bar{X}}$$

$$CI = [\bar{X} - ME_{CI}, \bar{X} + ME_{CI}]$$

$$ME_{PI} = t_{1-\frac{\alpha}{2}, \nu} s_X \sqrt{1 + 1/n}$$

$$PI = [\bar{X} - ME_{PI}, \bar{X} + ME_{PI}]$$

		α						
1-tailed		0.1	0.05	0.025	0.01	0.005	0.001	0.0005
2-tailed		0.2	0.1	0.05	0.02	0.01	0.002	0.001
	1	3.078	6.314	12.71	31.82	63.66	318.3	636.6
	2	1.886	2.920	4.303	6.965	9.925	22.33	31.60
	3	1.638	2.353	3.182	4.541	5.841	10.22	12.92
	4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
	5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
	6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
	7	1.415	1.895	2.365	2.998	3.499	4.785	5.408

Barley Yield



- From 5 samples, we find that $\bar{X} = 5.4 \frac{t}{ha}$
- From 5 samples, we find that $s = 0.08 \frac{t}{ha}$
- What is the 90% margin of error?

$$ME = t_{1-\frac{\alpha}{2}, \nu} s_{\bar{X}}$$
$$\nu = n - 1$$

From lecture 5, with

$$\sigma = 0.1 \frac{t}{ha}$$
$$ME_{CI} = 1.645 \times \frac{0.1}{\sqrt{5}}$$
$$= 0.074$$

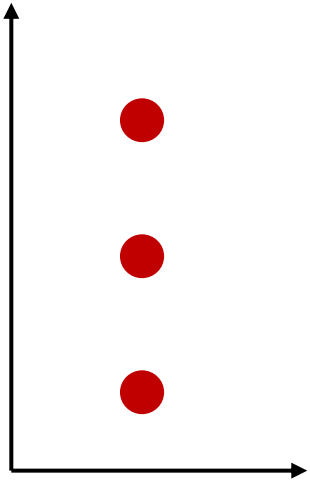
- Despite the randomly smaller s than σ , ME is larger for our sample than with the population values
- For small samples, the precision of our estimate is weaker

		α			
1-tailed		0.1	0.05	0.025	0.01
2-tailed		0.2	0.1	0.05	0.02
Degrees of Freedom, ν	1	3.078	6.314	12.71	31.82
	2	1.886	2.920	4.303	6.965
	3	1.638	2.353	3.182	4.541
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	5	1.476	2.015	2.571	3.365
	6	1.440	1.943	2.447	3.143
	7	1.415	1.895	2.365	2.998
	8	1.397	1.860	2.306	2.896
	9	1.383	1.833	2.262	2.821
	10	1.372	1.812	2.228	2.764
	11	1.363	1.796	2.201	2.718
	12	1.356	1.782	2.179	2.681
	13	1.350	1.771	2.160	2.650
	14	1.345	1.761	2.145	2.624
	15	1.341	1.753	2.131	2.602
	16	1.337	1.746	2.120	2.583
	17	1.333	1.740	2.110	2.567

<https://www.guinness.com/en-us/>

Intervals using the t-statistic and assumptions

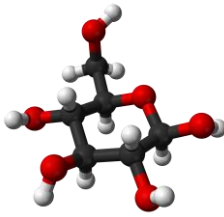
- If an RV is not normally distributed
- For confidence intervals,
 - If the sample is large enough
 - Central limit theorem tells us that the sample mean is approximately normally distributed with standard error $\sigma_{\bar{X}}$
 - We can use the estimate $s_{\bar{X}} \approx \sigma_{\bar{X}}$, hence $t_{1-\frac{\alpha}{2}, \nu} s_{\bar{X}}$ and the confidence interval
 - From 100 observations of $\bar{X}$, 95 of the CIs would contain μ
- For prediction intervals
 - Irrespective of sample size
 - s_X or σ_X will not be predictive of the probability under the PDF
 - If the PI was calculated using $t_{1-\frac{\alpha}{2}, \nu} s_X \sqrt{1 + 1/n}$
 - From 100 observations of X we cannot be sure how many would be in the PI



1 sample t-test

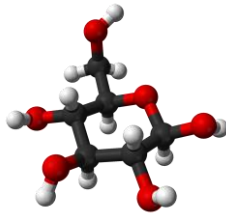
Analysis of one sample – in practice

Blood Glucose Levels



- For diabetics, the average blood glucose levels (mmol) can be modelled as normally distributed with a population mean of 5.5 mmol
- A group of 10 people have their blood glucose levels taken after trying a new low-carb diet, yielding a sample mean of 6.4 mmol and sample standard deviation of 1.2.
- With $\alpha = 0.01$, did the diet affect the average blood glucose level?

Blood Glucose Levels



$\alpha = 0.01$

$\mu_0 = 5.5 \text{ mmol}$

$n = 10$

$\bar{x} = 6.4 \text{ mmol}$

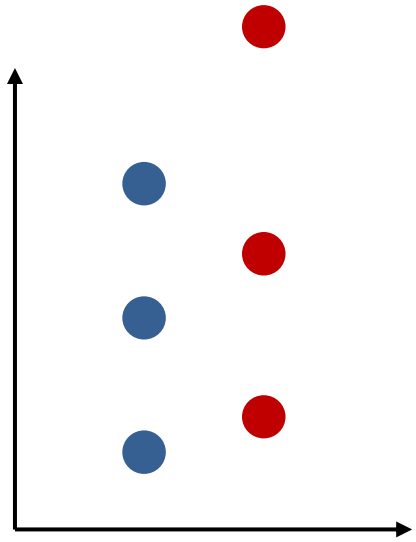
$s = 1.2 \text{ mmol}$

‘Affect’ doesn’t imply
direction: two-tailed

$H_0: \mu = \mu_0$

$H_1: \mu \neq \mu_0$

		α						
1-tailed		0.1	0.05	0.025	0.01	0.005	0.001	0.0005
2-tailed		0.2	0.1	0.05	0.02	0.01	0.002	0.001
Degrees of Freedom, ν	1	3.078	6.314	12.71	31.82	63.66	318.3	636.6
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	7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
	8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
	9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
	10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
	11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
	12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
	13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
	14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
	15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
	16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
	17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
	18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
	19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
	20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
	40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
	60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
	120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
	∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291



The independent two-sample t-test and CI

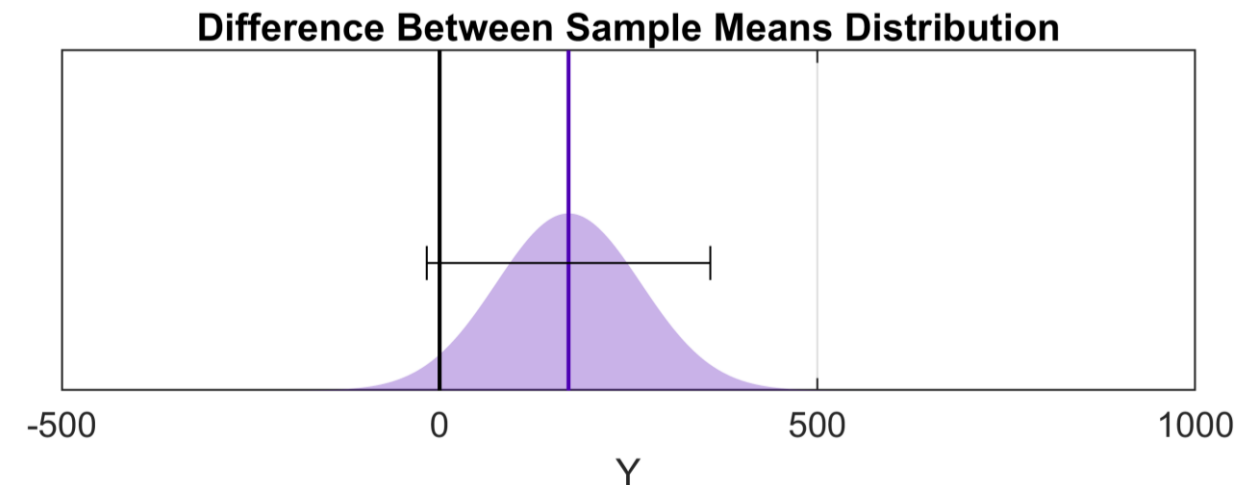
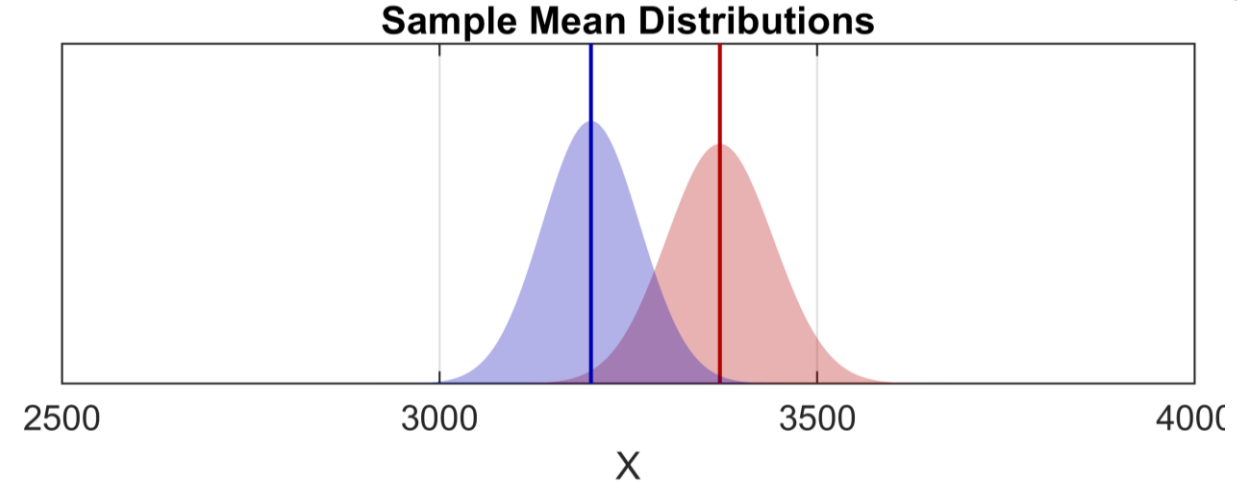
Analysing two unpaired samples – in practice

Dealing with two independent samples

- Sometimes, we wish to compare two samples, rather than just 1

$$\bar{Y} = \bar{X}_2 - \bar{X}_1$$
$$s_{\bar{Y}} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- Everything is as for the z-test, except for DOF



Welch-Satterthwaite Equation

- For 'similar variances'
 - typically considered to be $s_1^2 < 4 s_2^2$ or vice versa
 - The two terms contribute similarly to $s_{\bar{Y}}$
 - We have $\nu = n_1 + n_2 - 2$ degrees of freedom
- When the variances are not similar, we need to weight the DOF using the Welch-Satterthwaite equation:

Accounts for less information from one of the variances if much smaller than the other

$$\nu \approx \frac{\left(s_{\bar{X}_1}^2 + s_{\bar{X}_2}^2\right)^2}{\frac{s_{\bar{X}_1}^4}{n_1 - 1} + \frac{s_{\bar{X}_2}^4}{n_2 - 1}}$$

$$s_{\bar{X}_1}^2 = \frac{s_1^2}{n_1}$$

$$s_{\bar{X}_2}^2 = \frac{s_2^2}{n_2}$$

Pooled SD?

Some people use a pooled standard deviation, which is more complicated and while marginally more powerful for similar variances, is less powerful for different variances, so let's not use it!

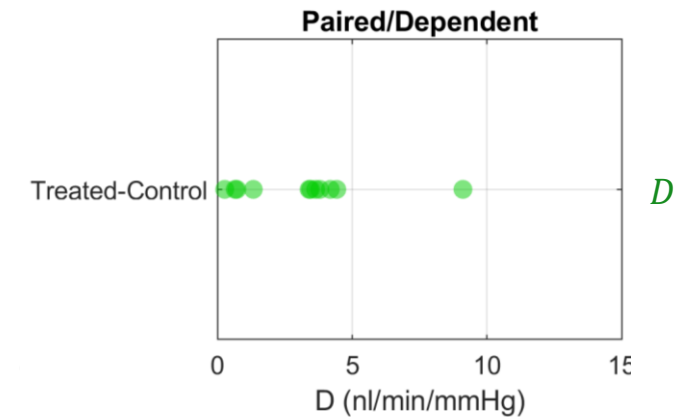
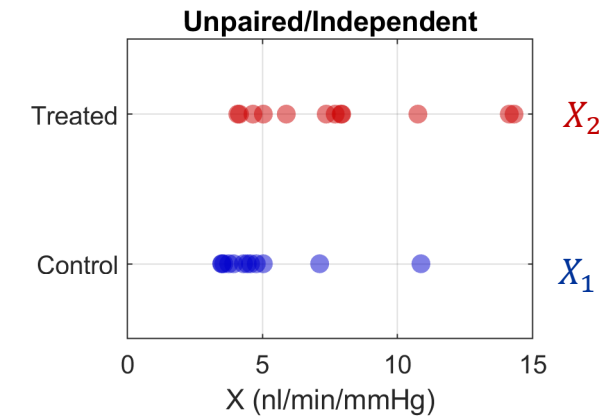
- For similar variances, this tends towards $n_1 + n_2 - 2$
- **I suggest always using the W-S, rather than relying on a threshold!**
- **Use this ν when calculating CI for difference between 2 independent samples**



- Experiments were carried out to evaluate the effect of PDA205 on the hydraulic conductance of the outflow pathways of the eye, C
- Previous studies have shown that two eyes from the same animal have similar C
- Hence, one eye was measured as a control, while the other was treated with PDA205
- The difference between the two eyes of each animal was analysed

$n_1 = 12$	$n_2 = 12$	$n_D = 12$
$\bar{x}_1 = 4.9$	$\bar{x}_2 = 7.8$	$\bar{d} = 2.9$
$s_1 = 2.1$	$s_2 = 3.6$	$s_D = 2.6$

- Here we ask: how would the conclusions differ if we considered the data as **unpaired**, vs **paired**?



Independent/Unpaired iPerfusion



$$n_1 = 12$$

$$n_2 = 12$$

$$\bar{x}_1 = 4.9$$

$$\bar{x}_2 = 7.8$$

$$s_1 = 2.1$$

$$s_2 = 3.6$$

Test statistic

H_0 : Difference of the means is zero

DOF - Is variance similar?

Difference between the means

Standard error on difference
between the means

Hence 'similar variance' solution
'should' be ok (but we generally avoid)

We'll compare to WS

Independent/Unpaired iPerfusion - DOF



$n_1 = 12$
 $\bar{x}_1 = 4.9$
 $s_1 = 2.1$

$n_2 = 12$
 $\bar{x}_2 = 7.8$
 $s_2 = 3.6$

$\bar{y} = 2.9$
 $s_{\bar{y}} = 1.20$

$t = 2.42$
 $\nu = ?$

```
>> tinv(1-0.05/2,17.7)

ans =

    2.1035
```

```
>> tinv(1-0.05/2,22)

ans =

    2.0739
```

		α			
	1-tailed	0.1	0.05	0.025	0
	2-tailed	0.2	0.1	0.05	0
Degrees of Freedom, ν	1	3.078	6.314	12.71	31
	2	1.886	2.920	4.303	6.
	3	1.638	2.353	3.182	4.
	4	1.533	2.132	2.776	3.
	5	1.476	2.015	2.571	3.
	6	1.440	1.943	2.447	3.
	7	1.415	1.895	2.365	2.
	8	1.397	1.860	2.306	2.
	9	1.383	1.833	2.262	2.
	10	1.372	1.812	2.228	2.
	11	1.363	1.796	2.201	2.
	12	1.356	1.782	2.179	2.
	13	1.350	1.771	2.160	2.
	14	1.345	1.761	2.145	2.
	15	1.341	1.753	2.131	2.
	16	1.337	1.746	2.120	2.
	17	1.333	1.740	2.110	2.
	18	1.330	1.734	2.101	2.
	19	1.328	1.729	2.093	2.
	20	1.325	1.725	2.086	2.
	40	1.303	1.684	2.021	2.
	60	1.296	1.671	2.000	2.
	120	1.289	1.658	1.980	2.
	∞	1.282	1.645	1.960	2.

- Can't read exact values from table
- Could draw conclusions anyway, or use interpolation or MATLAB
- In both cases, $|t| > t_{1-\alpha/2,\nu}$: we reject H_0



- Two-sample function in MATLAB is `ttest2`
- MATLAB needs to be told which assumption to use
 - See lines (126 and 127 'open `ttest2`')

- For similar variance assumption
 - Manually

`p_y=2*(1-(tcdf(2.40,22)))`; $p_y = 0.025$

- Using MATLAB in-built function

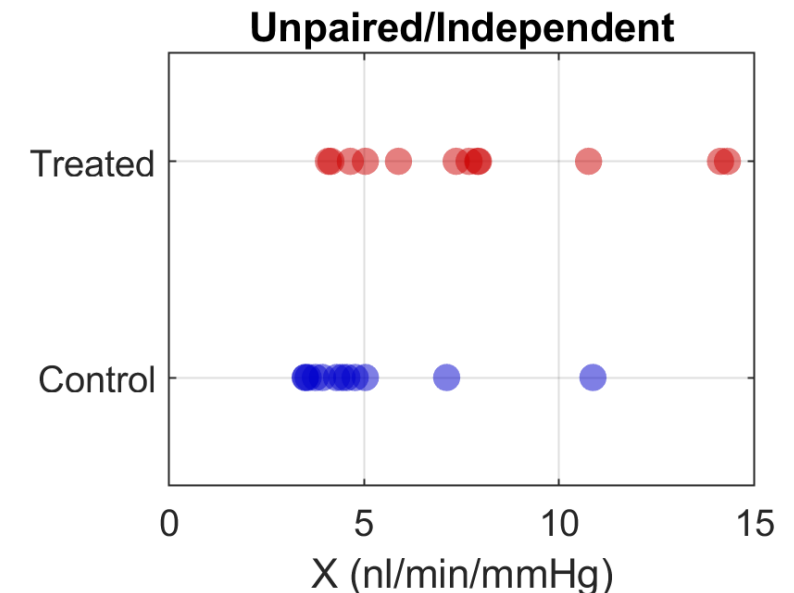
`[~,p]=ttest2(x2,x1,'vartype','equal')`

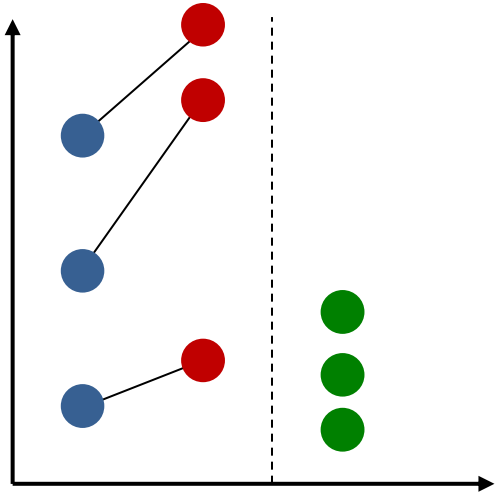
- For different variance assumption
 - Manually

`p_y=2*(1-(tcdf(2.40,17.7)))`; $p_y = 0.027$

- Using MATLAB in-built function
- `[~,p]=ttest2(x2,x1,'vartype','unequal')`
- But variances were 'similar'

- Both approaches find similar values for this data
- Difference will be more important for small sample sizes
- **I advise just always using 'unequal'/W-S by default**
- Less reliance on thresholds!





The dependent two-sample t-test and CI

Analysing two paired samples - in practice

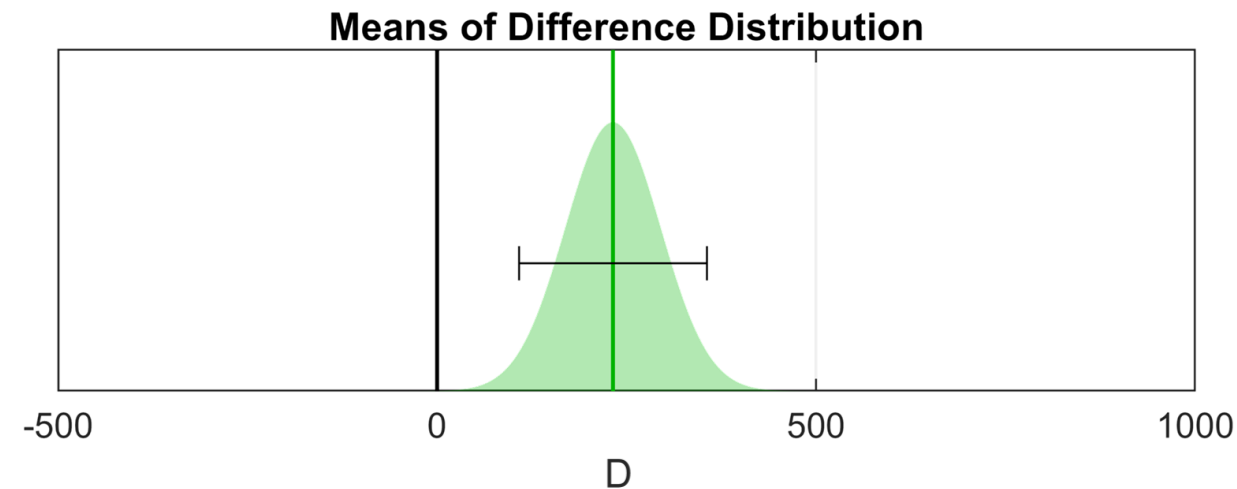
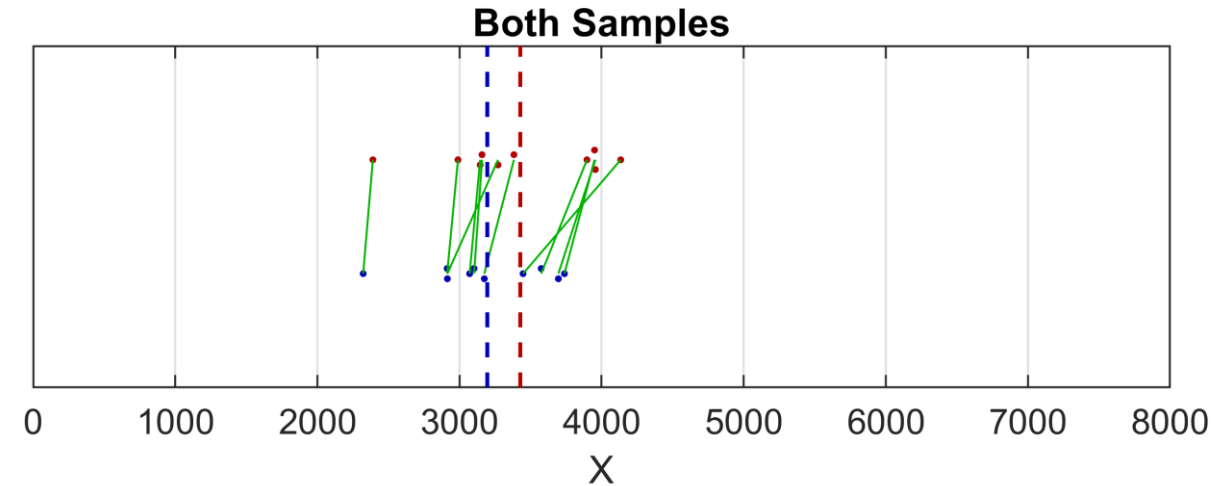
Dealing with two dependent samples

- Where possible, we use paired designs to reduce the effects variability between individuals
- Everything is as for the z-test, and DOF is easy, as we just have n samples

$$\bar{D} = \overline{X_2 - X_1}$$

$$s_{\bar{D}} = \frac{s_D}{\sqrt{n}}$$

$$\nu = n - 1$$

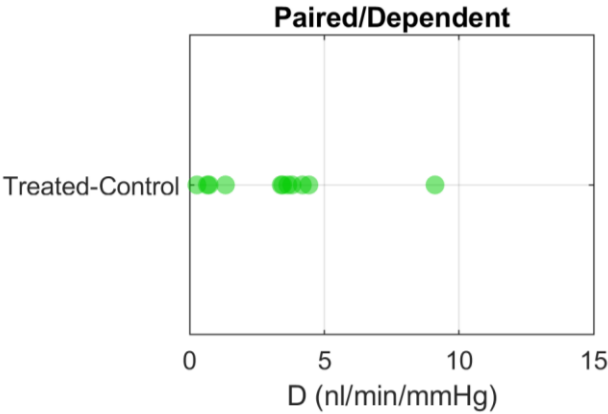


iPerfusion Dependent/Paired



$n_D = 12$
 $\bar{d} = 2.9$
 $s_D = 2.6$

H_0 : Mean of the differences is zero



		α			
1-tailed		0.1	0.05	0.025	0.01
2-tailed		0.2	0.1	0.05	0.02
Degrees of Freedom, ν	1	3.078	6.314	12.71	31.82
	2	1.886	2.920	4.303	6.965
	3	1.638	2.353	3.182	4.541
	4	1.533	2.132	2.776	3.747
	5	1.476	2.015	2.571	3.478
	6	1.440	1.943	2.447	3.317
	7	1.415	1.895	2.365	3.217
	8	1.397	1.860	2.306	3.143
	9	1.383	1.833	2.262	3.098
	10	1.372	1.812	2.228	3.055
	11	1.363	1.796	2.201	3.024
	12	1.356	1.782	2.179	3.000
	13	1.350	1.771	2.160	2.980
	14	1.345	1.761	2.145	2.963
	15	1.341	1.753	2.131	2.950
	16	1.337	1.746	2.120	2.940
	17	1.333	1.740	2.110	2.932
	18	1.330	1.734	2.101	2.925
	19	1.328	1.729	2.093	2.919
	20	1.325	1.725	2.086	2.914
40	1.303	1.684	2.021	2.851	
60	1.296	1.671	2.000	2.833	
120	1.289	1.658	1.980	2.819	
∞	1.282	1.645	1.960	2.807	



- One sample or paired dependent t-test function in MATLAB is

`ttest`

- MATLAB will deal with either x_1 and x_2 or d

- Manually

```
pd=2*(1-tcdf(3.84,11));
```

- Using MATLAB in-built function for separate groups

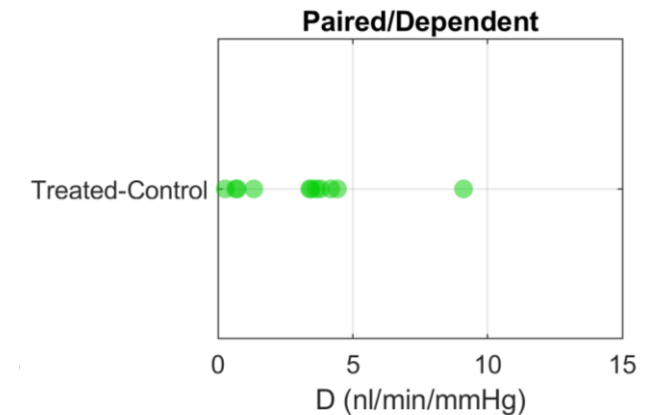
```
[~,p]=ttest(x2,x1)
```

- Using MATLAB in-built function for a one sample t-test groups

```
d=x2-x1;
```

```
[~,p]=ttest(d);
```

$$p_d = 0.0028$$



Paired vs Unpaired



- **Independent/unpaired**

$$p_Y = 0.027$$

- **Dependent/paired**

$$p_D = 0.0028$$

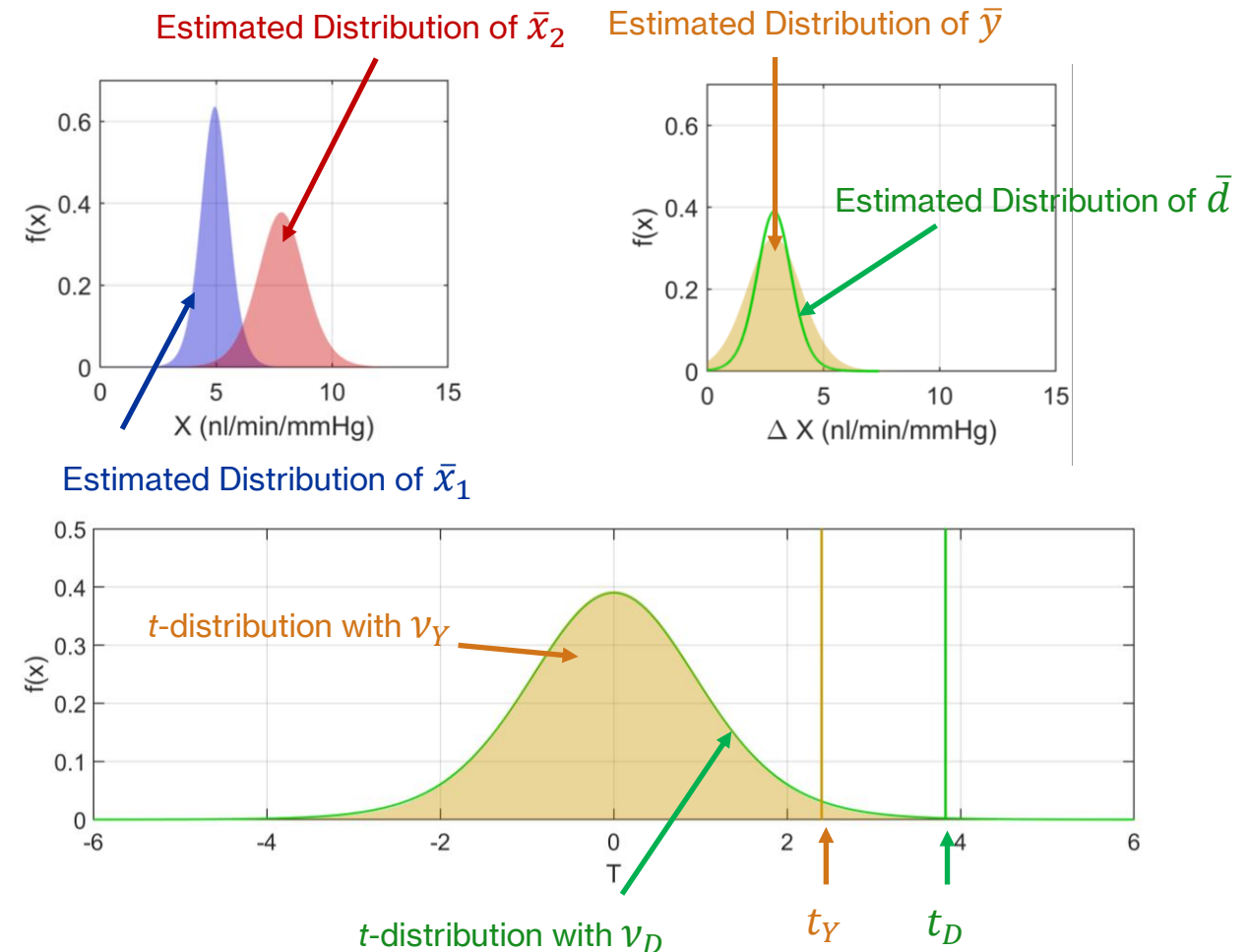
- Comparison is meaningful because these are the same data!

- Although

- $\nu_Y \gg \nu_D$
- $t_D \gg t_Y$

- Note, paired test would always be used in this situation!

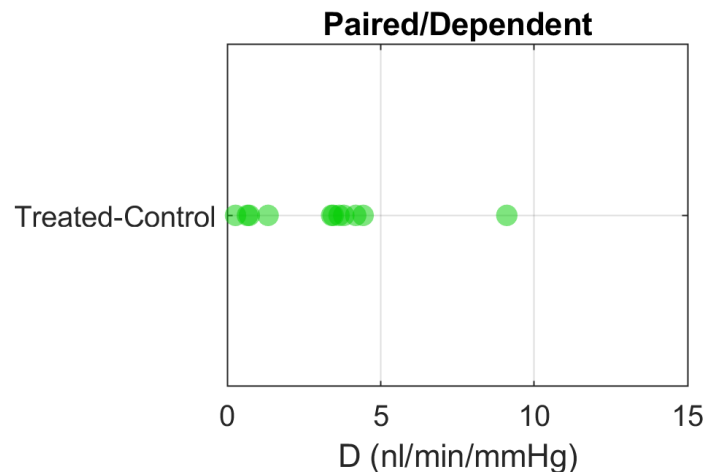
- (This is just a demonstration)





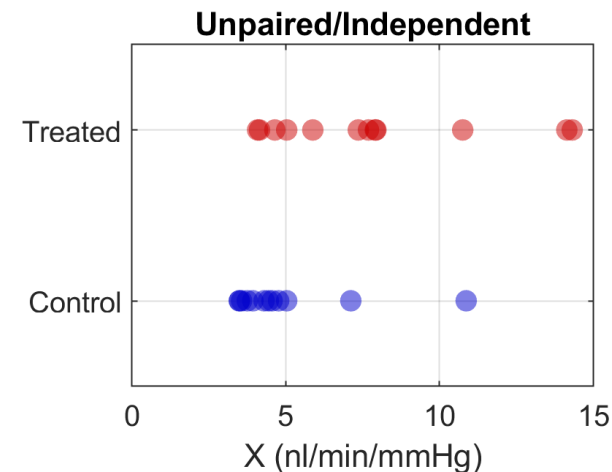
Using paired analysis

- The average difference in hydraulic conductance between in the treated and the contralateral control eyes was 2.9 [1.2, 4.5] nl/min/mmHg ($p < 0.01$, $n = 12$)



Using unpaired analysis

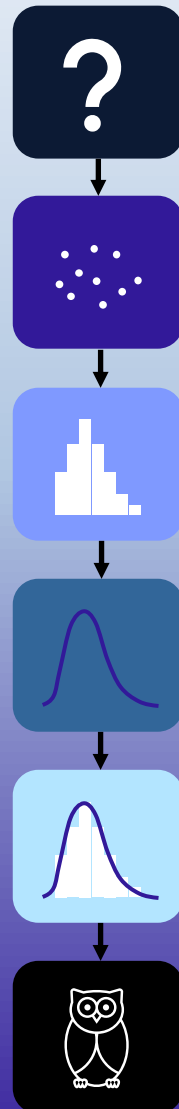
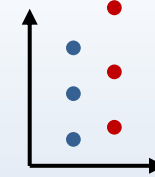
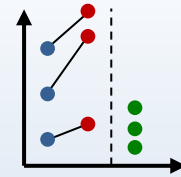
- The average hydraulic conductance in treated eyes was 2.9 [0.4, 5.4] nl/min/mmHg higher than in contralateral control eyes ($p < 0.05$, $n_1 = 12$, $n_2 = 12$)





Summary of t-tests

for reference and understanding



1-sample t-test	Paired 2-sample t-test	Unpaired 2-sample t-test
How does the ...		
population average μ_X	population average difference μ_D	difference between the averages of the two populations, μ_1 and μ_2
... compare to some known value μ_0 ?		
Capture $i = 1:n$ observations, x_i	Capture $i = 1:n$ paired observations, $x_{1,i}$ & $x_{2,i}$ to yield $d_i = x_{2,i} - x_{1,i}$	Capture $i = 1:n_1$ & $j = 1:n_2$ unpaired observations, $x_{1,i}$ & $x_{2,j}$
Visualisation of x_i	Visualisation of $x_{1,i}$ & $x_{2,i}$ and/or d_i	Visualisation of $x_{1,i}$ & $x_{2,j}$
x_i	d_i	$x_{1,i}$ and $x_{2,j}$
... are sampled from (a) normally distributed population(s) with unknown standard deviation(s) ...		
s_X	s_D	s_1 and s_2
$s_{\bar{X}} = s_X / \sqrt{n}$ $t = \frac{\bar{x} - \mu_0}{s_{\bar{X}}}$ $v_X = n - 1$	$s_{\bar{D}} = s_D / \sqrt{n}$ $t = \frac{\bar{d} - \mu_0}{s_{\bar{D}}}$ $v_D = n - 1$	$\bar{y} = \bar{x}_2 - \bar{x}_1, s_{\bar{X}_1} = s_{X_1} / \sqrt{n_1}, s_{\bar{X}_2} = \frac{s_{X_2}}{\sqrt{n_2}},$ $s_{\bar{Y}} = \sqrt{s_{\bar{X}_1}^2 + s_{\bar{X}_2}^2}, t = \frac{\bar{y} - \mu_0}{s_{\bar{Y}}}$ $v_Y \approx \left(s_{\bar{X}_1}^2 + s_{\bar{X}_2}^2 \right)^2 / \left(\frac{s_{\bar{X}_1}^4}{n_1 - 1} + \frac{s_{\bar{X}_2}^4}{n_2 - 1} \right)$
$H_0: \mu_X$ or μ_D or $\mu_2 - \mu_1$ is not 'different to'/'greater than'/'less than' μ_0 (two-/upper-/lower-tailed) Is observed t likely if H_0 is true, for a given α ? {yes/no}		
We found a mean difference of $\bar{x}$ or $\bar{d}$ or $\bar{y}$ [95% CI] that {was/was not} statistically different from μ_0		

Sample size and power

- No analytical solutions for t-test. Instead, use MATLAB

'z' or 't' or 't2'

z-test

t-test

Sample size '1-Beta, []'

Power '[], n'

'both' or 'right' or 'left'

2-tailed

upper-
tailed

lower-
tailed

sampsizepwr('z', $[\mu_0 \text{ } \sigma \text{ or } s]$, μ_1 , $1 - \beta$, n , 'alpha', α , 'tail', 'both')

- For 1 sample (z or t)
 - $[\mu_0, \sigma_X \text{ or } s_X]$ and $[\mu_X]$
- For paired data (z or t)
 - $[\mu_0, \sigma_D \text{ or } s_D]$ and $[\mu_D]$
- For unpaired data with known variance (z-test) (z)
 - $[\mu_0, \sigma_Y]$ and $[\mu_Y]$
- For unpaired data with unknown variance (t-test) (t2)
 - $[\mu_1, s_1 = s_2]$ and $[\mu_2]$

Note, for independent t-test, MATLAB assumes **equal variances** and uses the **pooled standard deviation**

This is spherical cow.

“Originating in theoretical physics, the metaphor refers to physicists' tendency to reduce a problem to the simplest form imaginable in order to make calculations more feasible, even if the simplification hinders the model's application to reality.”

Wikipedia

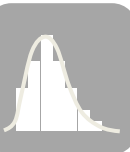
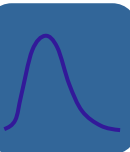
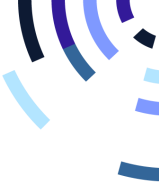


Assumptions

Required for using statistics!

Assumptions

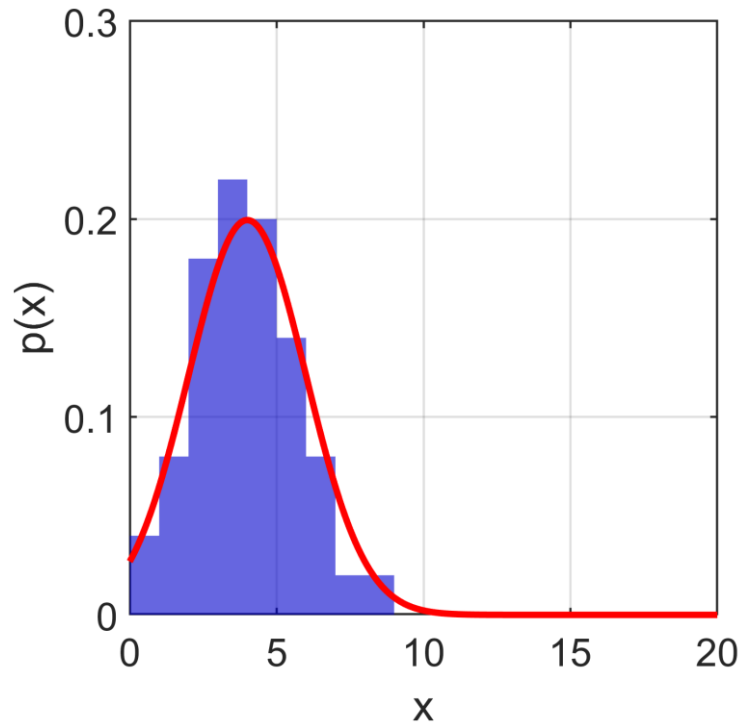
- For the z-tests and t-tests, we've made the **assumption** that the **sample means** were **normally distributed**
- As we saw in Lecture 5, this is true if
 - The underlying data are sampled from a normal distribution and/or
 - There is a large sample size such that CLT tells us that the sample means are normally distributed, even for non-normal underlying data
- **But**, the sample size required for this to hold depends on the specific data (see exercises)
- These approaches are 'parametric'
 - They make underlying assumptions about the distribution from which the sample is taken
 - They use estimators of the **parameters** of the distribution (μ and σ^2/n)
- If correct, this allows us to make better 'guesses' about the underlying populations
- If this is not the case, then the statistics we derived can be incorrect
- This could increase the Type I or Type II error rate in poorly defined ways
- We need to be able to test if the underlying distribution is what we think it is
- If it is not, we need non-parametric statistics that do not make assumptions about the underlying distribution (Lecture 8)



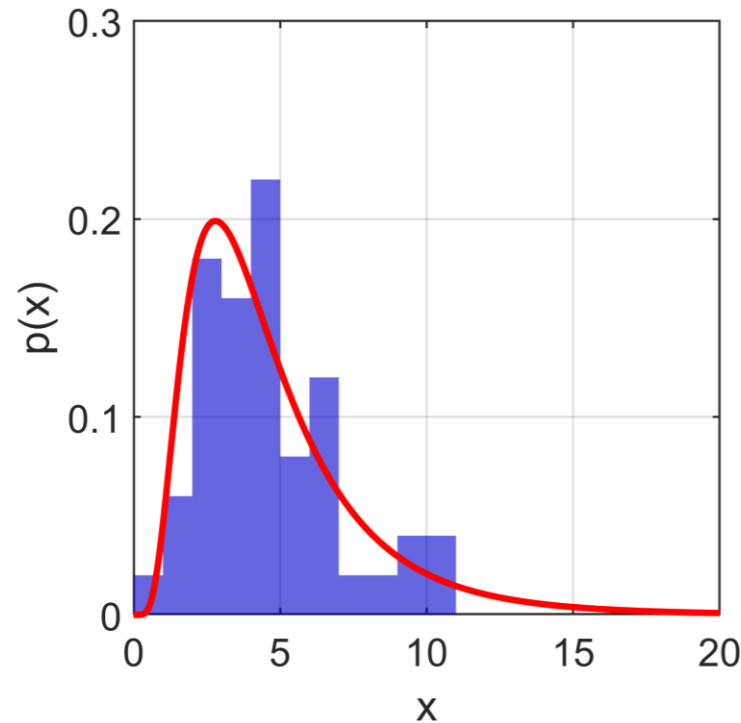
Visualising Different Distributions



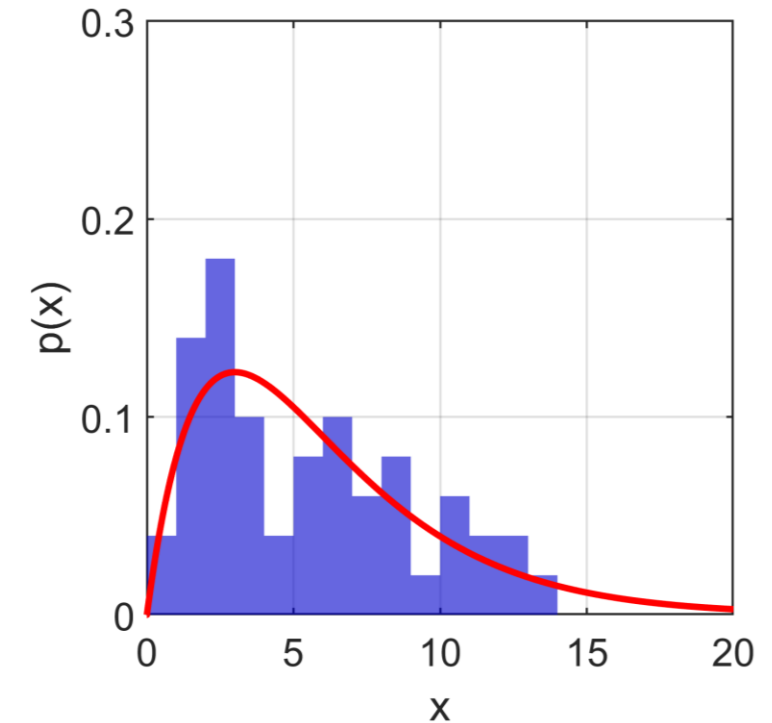
Normal



Lognormal



Gamma



- Red lines show underlying population (we won't know)
- Histograms roughly show us how the distribution looks
- It takes some interpretation, which could be subjective
- We need better ways to compare distributions

Introduction to Empirical CDF

- For RVs with known probability distributions, we can define the CDF

$$F(x) = P(X \leq x) = \sum_{y: y \leq x} p(y)$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

- For a finite data set X_i we can calculate an 'empirical' CDF, using

All the numbers in the data set that are less than x



$$F_E(x) = \frac{1}{n} \sum_{i=1}^{X_i \leq x} 1$$

```
i=1:n;
FE=i./n;
```

- Sort X_i
 - For each value x of X_i
 - $F_E(x)$ is the proportion of values less than that number
 - Maximum value will equal 1
- E.g. $x=[1, 4, 5, 7, 9]$
 - $F_E(7) = 0.8$
 - If you randomly selected an element from X_i , there is an 80% probability it would be below 7

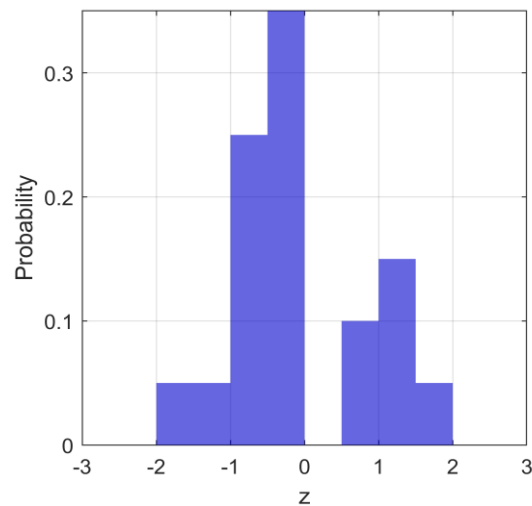
Empirical CDF

$$F_E(z) = \frac{1}{n} \sum_{i=1}^{Z_i \leq z} 1$$

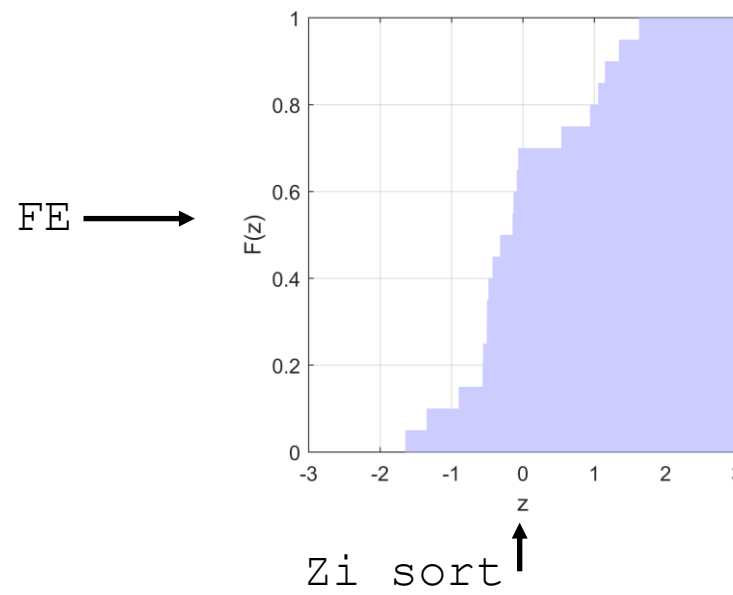
Example with standard normal distribution

```
Zi=randn(20,1);           % generate some standard random normal data
Zi_sort=sort(Zi);          % sort the data
n=numel(Zi);               % get the number of data points
i=1:n                      % array of indices
FE=i./n;                   % Values for the ECDF
```

Histogram



ECDF



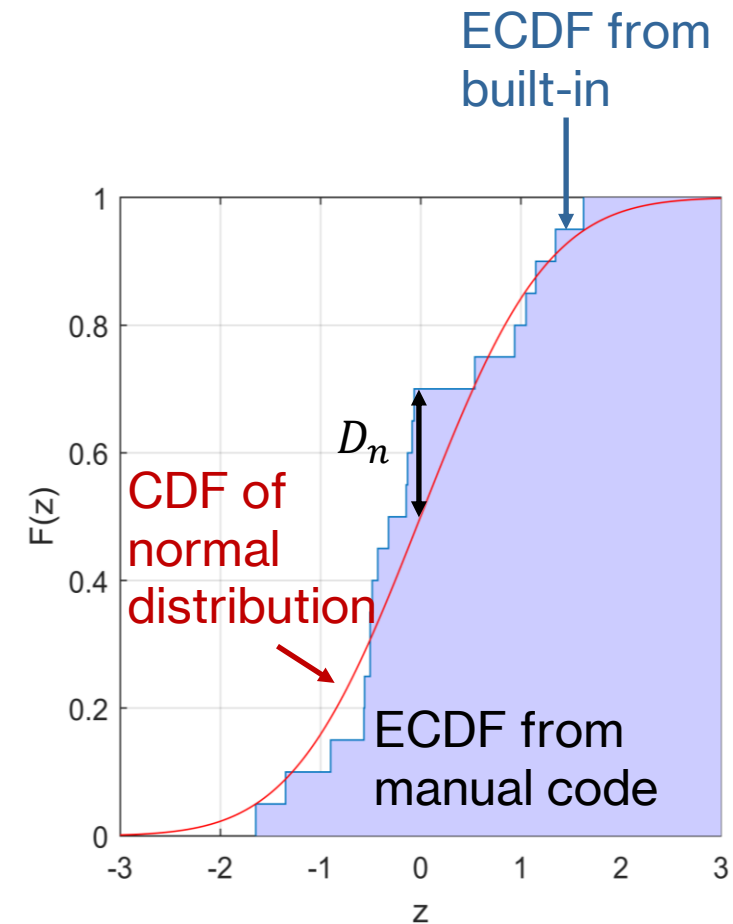
```
[Fm, xm]=ecdf(Zi)
% in-built MATLAB
function
```

See Lecture7_code for plotting

Using the Empirical CDF

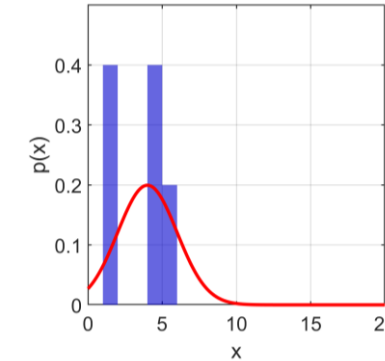
$$F_E(x) = \frac{1}{n} \sum_{i=1}^{X_i \leq x} 1$$

- By comparing the normal CDF to the empirical CDF, we can visualise whether the two distributions are similar
- The **Kolmogorov-Smirnov test** (`kstest`) finds the maximum value of the difference between the ECDF and CDF of interest
 - typically normal
 - could also be another distribution
 - including another empirical CDF if you want to compare two samples
- Creates a test statistic $K = \sqrt{n}D_n$
 - note, the n data points must be independent
- Calculates a p -value for the null hypothesis that the data come from the proposed underlying distribution
- **Anderson-Darling test** (`adtest`) works on a similar principle, but is more powerful for smaller samples



Introduction to Q-Q plots

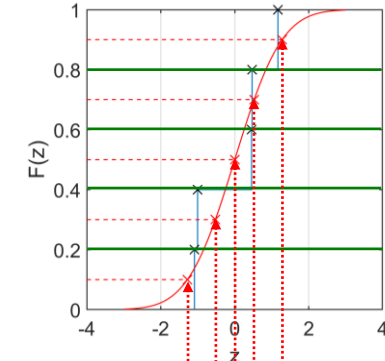
- Consider random 5 data points samples from a normal distribution
- Let's standardise and plot the **ECDF** and Φ
- Quantiles divide probability distribution into n sections of equal probability
- Evaluate the **quantile at the middle probability of each section**
 - 0.1, 0.3, 0.5, 0.7, 0.9
- z-values from normal distribution go on **x axis**
- Sample z-values go on y-axis
- For perfect normal distribution, all data points would lie on line of unity



e.g. 1st point

$$\Phi^{-1}(0.1) = -1.28$$

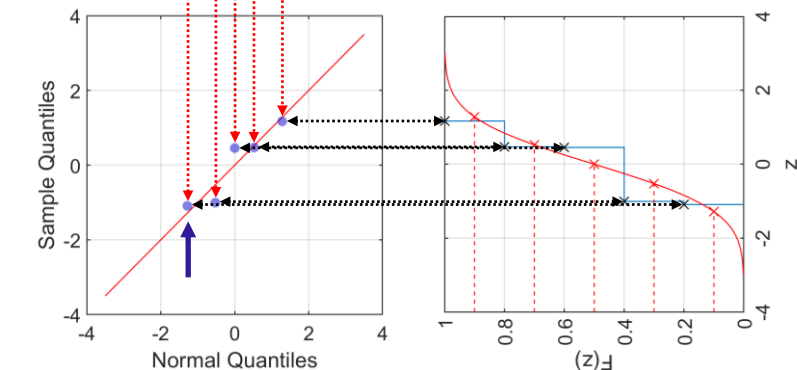
If we had a normal distribution, the expected z-value of the 1st of 5 data points would be -1.28 (i.e. 1.28 standard deviations less than the mean)



Lowest standardised data point is -1.09 (i.e. 1.09 standard deviations less than the mean)

So closer to the mean than expected for a normal distribution

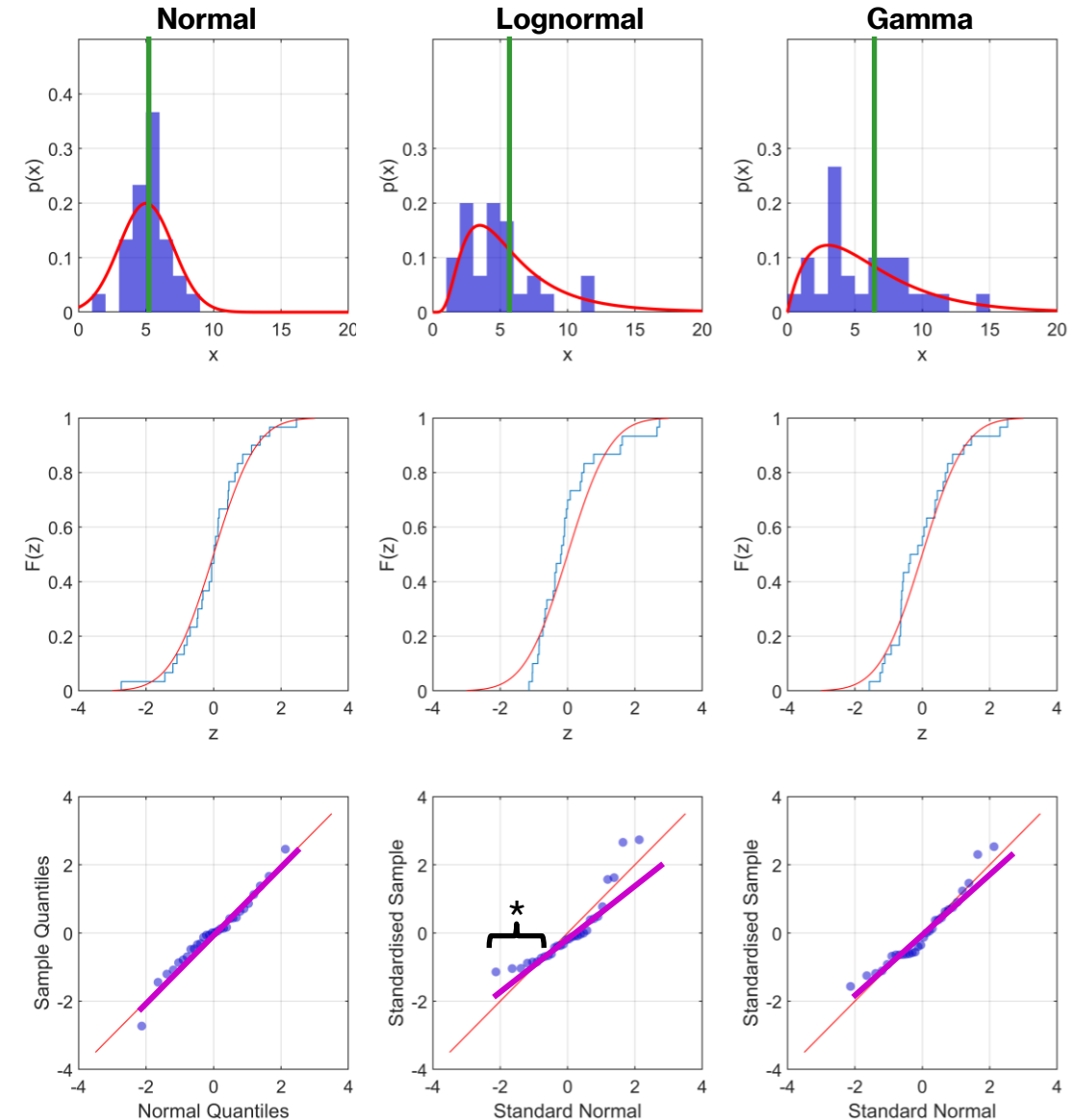
z values from black crosses



z values from red crosses via $\Phi^{-1}F(z)$

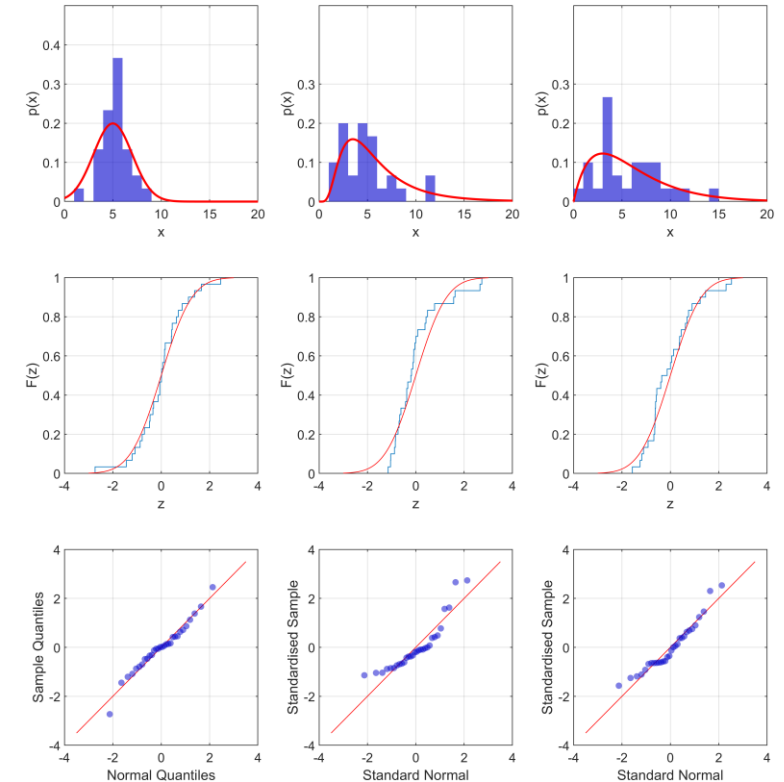
QQ plots and the Shapiro-Wilk Test

- For normal data, data points lie mostly on line of unity
- Implies sample distribution is consistent with coming from an underlying normal distribution
- **Shapiro-Wilk test** (`swtest`) calculates a test statistic based on the **slope** of the Q-Q plot
- The most powerful test for normality for smaller samples
- Note e.g. *
 - Sample quantiles are smaller magnitude than expected for normal
 - more values than expected below the mean if the data were normally distributed



Summary of Normality Criterion

- The NHST tests we have looked at so far require that the sample means are normally distributed
- If we don't have 'large' samples, this will be safe if all groups are sampled from normal distributions
- KS, AD and SW tests are ways to check this
 - H_0 = the sample comes from a population that follows a normal distribution
 - H_1 = the sample comes from a population that does not follow a normal distribution
- If the null-hypothesis is rejected, then the NHST tests may not be accurate because the model is not accurate
 - α and β may be larger than expected
 - We should trust conclusions even less than normal!



$$p_{ks} = 0.89$$

$$p_{ad} = 0.51$$

$$p_{sw} = 0.29$$

Data are normal
 H_0 not rejected
 TN

$$p_{ks} = 0.14$$

$$p_{ad} = 0.0005$$

$$p_{sw} = 0.0008$$

Data are not normal
 H_0 rejected
 TP

$$p_{ks} = 0.50$$

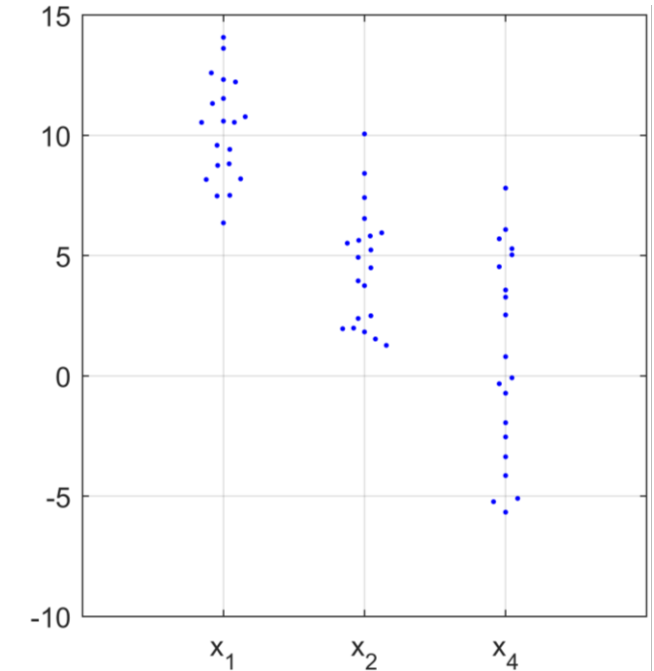
$$p_{ad} = 0.08$$

$$p_{sw} = 0.07$$

Data are not normal
 H_0 not rejected
 FN

Equality of Variance

- For Pearson's Correlation Coefficient (Lecture 8) and ANOVA (Lecture 9), we assume that samples come from distributions with equal variance
- This property is also known as homoscedasticity
- Ho-mo-sid-ass-tiss-it-ee
- We won't go into how these tests work in this course
- Note that for t -test, W-S correction to DOF means this isn't an issue
- Two common variance tests are
 - Bartlett's Test
 - More powerful for normal data
 - Levene's Test
 - More robust for non-normal data



Equality of Variance

Simulate 4 normal distributions x_1, x_2, x_3, x_4 with variances

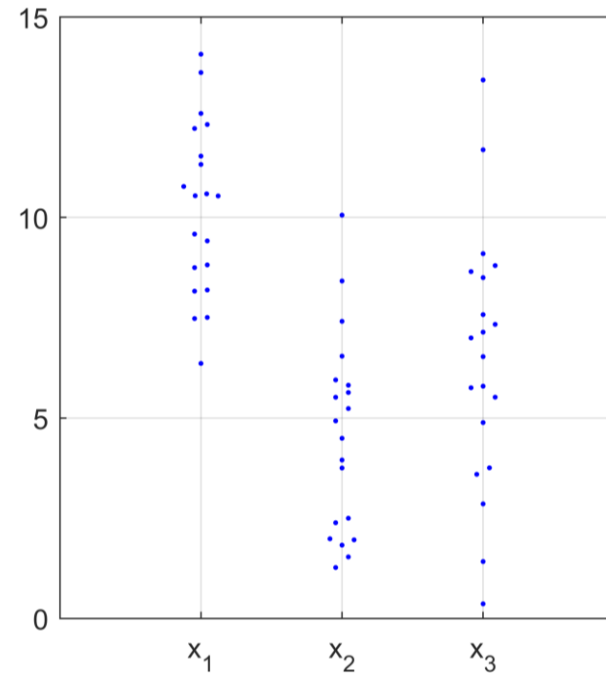
```
v1=5; v2=6; v3=7; v4=15;
```

Compare x_1, x_2, x_3

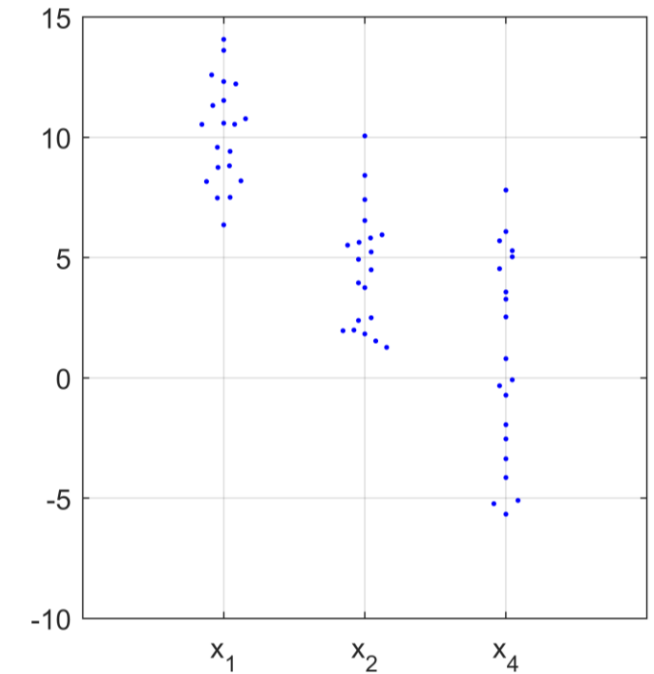
```
pBart=varstestn([x1 x2 x3],  
'testtype', 'Bartlett')  
pLev=varstestn([x1 x2 x3],  
'testtype', 'LeveneAbsolute')
```

Compare x_1, x_2, x_4

```
pBart=varstestn([x1 x2 x4],  
'testtype', 'Bartlett')  
pLev=varstestn([x1 x2 x4],  
'testtype', 'LeveneAbsolute')
```



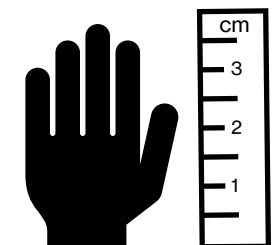
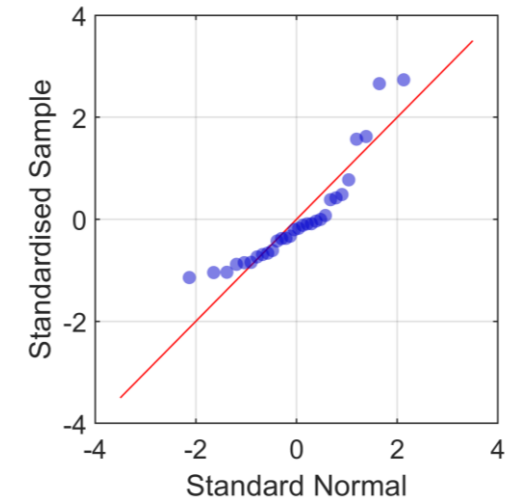
$$p_B = 0.2, p_L = 0.4$$



$$p_B = 0.005, p_L = 0.0003$$

When normality or homoscedasticity are violated

- If n is not 'large'
 - **and** the assumption of normality is violated (for z-test, t-test, ANOVA, Pearson's Correlation Coefficient)
 - **or** homoscedasticity is violated (for Pearson's Correlation Coefficient and ANOVA)
- Investigate transformation
 - Modify the data such required assumptions are not rejected
- Use non-parametric tests
 - Analogous to Pearsons, t -tests and ANOVA, but focus on ranks rather than values
- There are also a branch of tests called 'robust tests' that are more robust to outliers and violations of assumptions
 - See Field, Miles and Field 'Discovering Statistics Using ...' for more information if you are interested
- **Note on shorthand**
 - We tend to do tests, then conclude (colloquially)
'X is normally distributed'
 - However, we never know the truth, so really we mean
'The data are consistent with having been sampled from a normal distribution'
 - It is a pain to say this every time, so the shorthand is ok
 - in the 'real world' but be precise in exams and coursework
 - Could say '*not not normally distributed*'!





Transformation and the lognormal distribution

Defining a new RV that might comply with assumptions

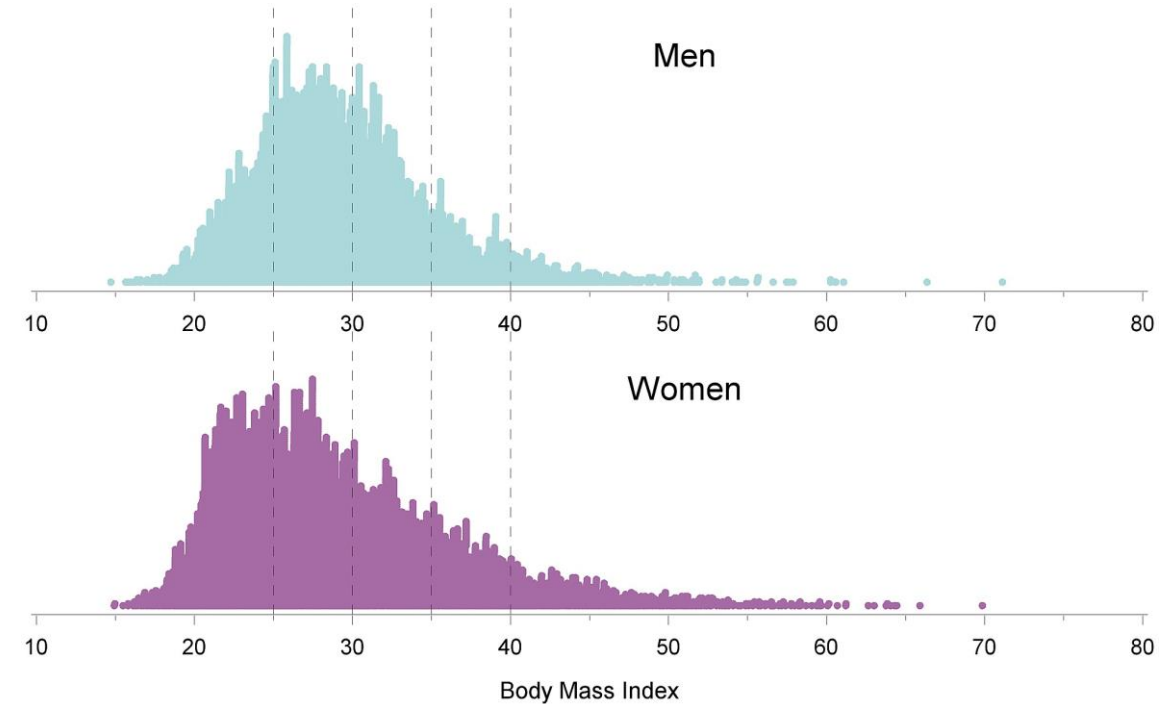
Why transform the data?



- Take Body Mass Index (BMI) as an example

$$BMI = \frac{Weight}{Height^2}$$

- Weight and height are approximately normally distributed in the population (for men or for women)
- Normal distributions are additive
- The ratio of two normal distributions is not a normal distribution
- One would not expect a normal distribution for BMI!
- Transformation may help us make a new normally distributed variable $Y=f(BMI)$



[Feng et al.](#)

BMI and transformation



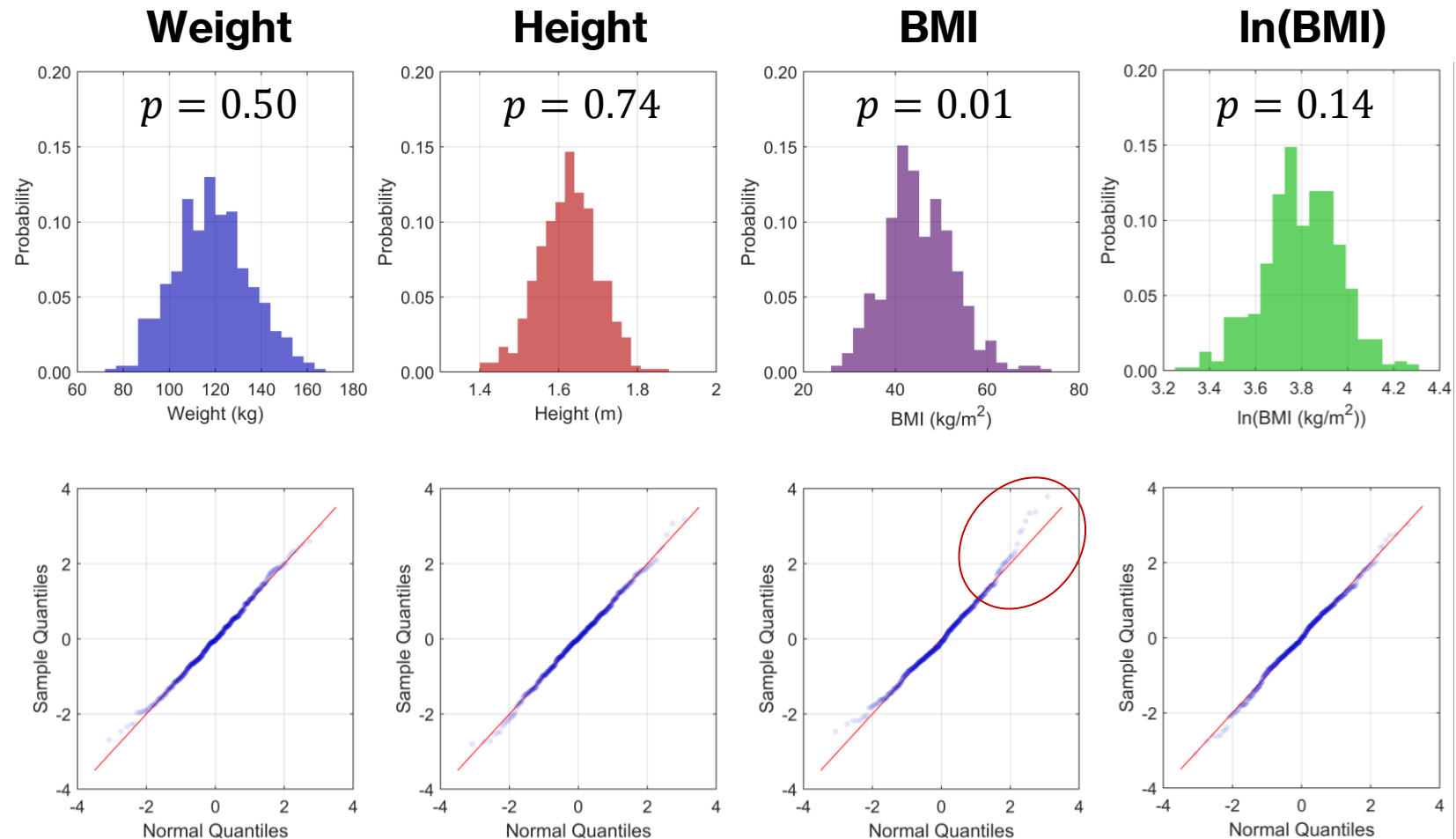
- From [Feng et al.](#): Females from a biobank of morbidly obese individuals
- $W \sim N(120, 17^2)$ kg, $H \sim N(1.62, 0.08^2)$ m, $N = 477$

- Use `adtest`
- H_0 rejected for BMI
- Try log transform

$$Y = \ln(BMI)$$
- H_0 not rejected for $\ln(BMI)$

$$Y \sim N(\mu_Y, \sigma_Y^2)$$

- We can now use e.g. t-test etc. (large sample size would have been ok anyway)



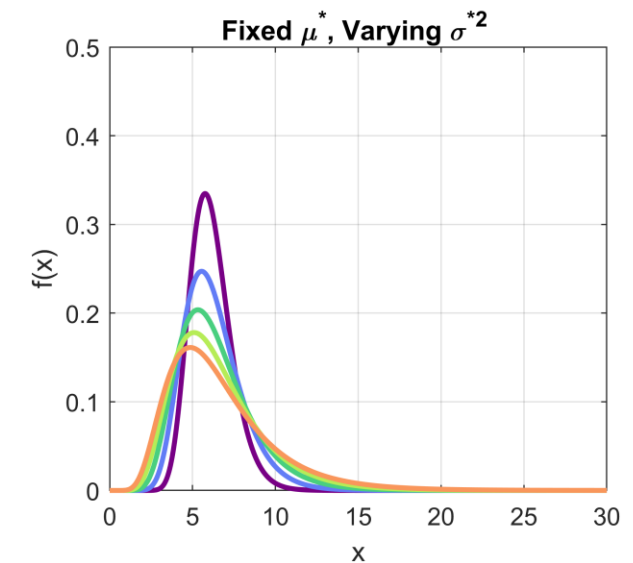
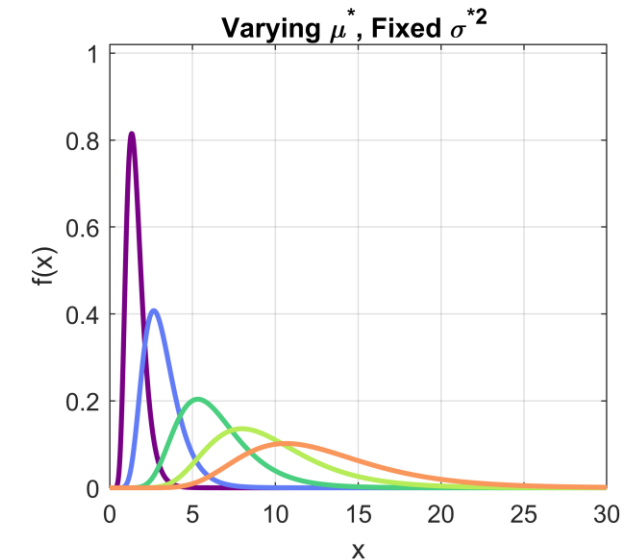
Introduction to the Lognormal Distribution

- It is quite common to take the logarithm of data and find the outcome to be consistent with being sampled from a normal distribution
 - shorthand: being normally distributed
- In general, if $Y = \ln(X) \sim N(\mu_Y, \sigma_Y^2)$

$$X \sim \Lambda(\mu_X^*, \sigma_X^{*2})$$

- X is 'lognormally distributed' with parameters μ_X^* and σ_X^{*2}
- In general, quite a lot of data are actually inherently log-normally distributed (read [Limpert et al. 2001](#))
- In particular
 - 'multiplicative' data such as BMI

$$BMI = W/H^2 \longrightarrow \ln(BMI) = \ln(W) - 2 \ln(H)$$

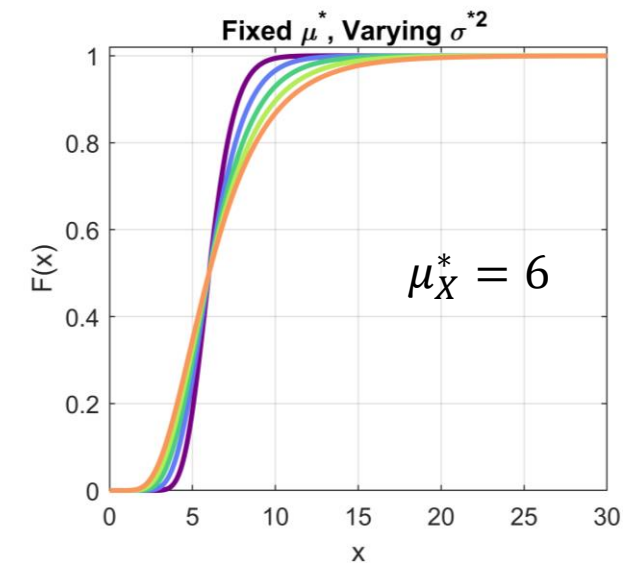
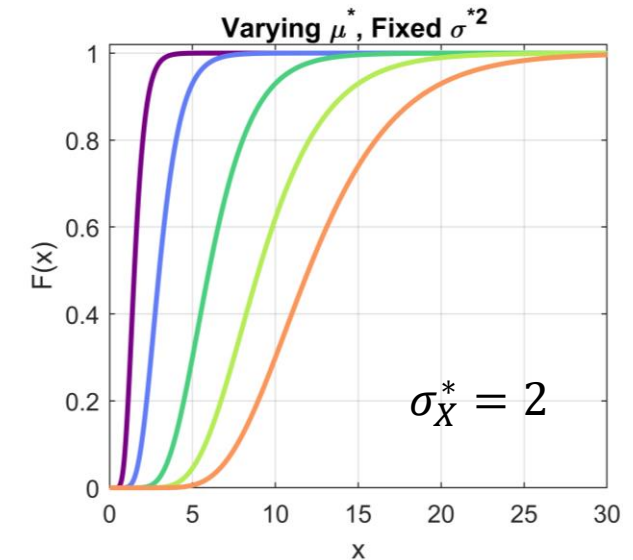


Lognormal Parameters

$$Y = \ln(X) = N(\mu_Y, \sigma_Y^2)$$

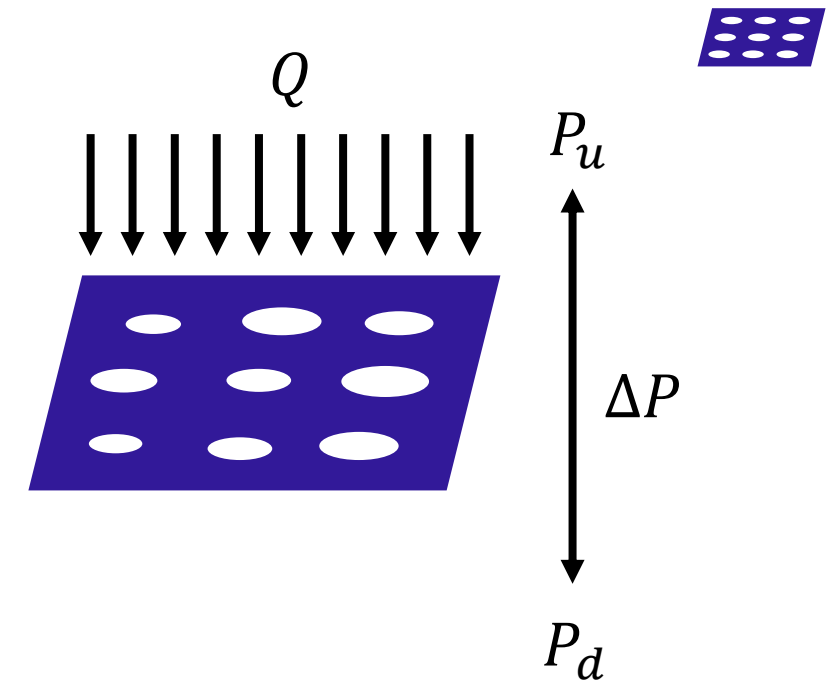
$$X \sim \Lambda(\mu_X^*, \sigma_X^{*2})$$

- μ_X^* is the geometric population mean
 - $\ln(\mu_X^*) = \mu_Y$ or equivalently $\mu_X^* = e^{\mu_Y}$
- The sample geometric mean is $\bar{x}^*$
 - $\ln(\bar{x}^*) = \bar{y}$ or equivalently $\bar{x}^* = e^{\bar{y}}$
- σ_X^* is the geometric population standard deviation
 - $\ln(\sigma_X^*) = \sigma_Y$ or equivalently $\sigma_X^* = e^{\sigma_Y}$
- The sample geometric standard deviation is s_X^*
 - $\ln(s_X^*) = s_Y$ or equivalently $s_X^* = e^{s_Y}$
 - $\ln(s_X^{*2}) = 2s_Y$ or equivalently $s_X^{*2} = e^{2s_Y}$
- Log-transform data, do stats, then take exponential (antilog) for mean, CI etc.
 - hence not on formula sheet!



Conductivity

- The hydraulic conductivity of a membranes, $C=Q/\Delta P$ is measured, yielding the following values.
- Given that the data are sampled from a lognormal distribution, find the 95% confidence interval
 $C = [0.07, 0.09, 0.08, 0.24]$

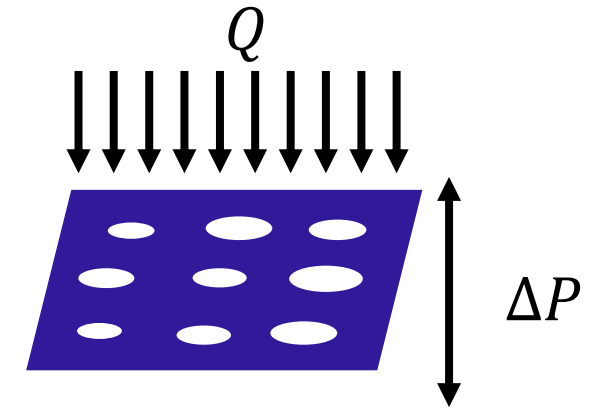


More properties of lognormal distribution

- If Y_1 and Y_2 are both lognormally distributed, then the ratio Y_2/Y_1 is also lognormally distributed

$$\ln\left(\frac{Y_2}{Y_1}\right) = \ln(Y_2) - \ln(Y_1) = X_2 - X_1$$

- This is useful when we are interested in a fold increase
 - E.g. Conductance was 1.1-fold greater on average with a new method to make membranes
 - We can now calculate confidence intervals on the fold increase 1.10 [1.05, 1.20] fold
 - Or - 1 and $\times 100\%$ so we can report a 10 [5, 20] % increase
- Cannot do this if Y_1 and Y_2 are normally distributed
- If data cannot be negative (e.g. conductivity, length, mass)
 - e.g. for $\bar{x} = 6, s = 6$
 - 16% ($\Phi^{-1}(-1)$) of observations would be negative
 - keep an eye out for SEM, e.g. for $n = 9$ someone might report
 - 6 ± 2 [mean $\pm$ SEM]....
- Is indistinguishable from a normal distribution when $\mu \gg \sigma$



Conductance = 1/ Resistance
 $C = 1/R$

$$\ln(C) = \ln\left(\frac{Q}{\Delta P}\right) = \ln(Q) - \ln(P)$$

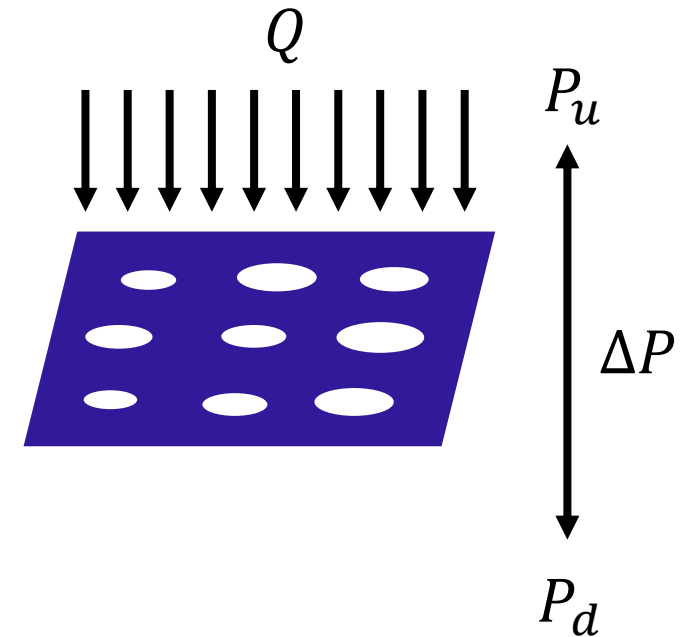
$$\ln(R) = \ln\left(\frac{P}{\Delta Q}\right) = \ln(P) - \ln(Q)$$

$$\ln(C_2/C_1) = \ln(C_2) - \ln(C_1)$$

These work if $\mu \gg \sigma$ for Q and P,
 otherwise some values of C or R will
 tend to zero or infinity

Physics-based Transformation

- Other options include
 - Reciprocal
 - Square
 - Square root
- It is not clear whether these fixes actually improve the power
 - they probably do for some cases and not for others
- In general, consider the physics/mathematics
- For example conductance ($C=Q/\Delta P$) of flow through a porous membrane with variable size pores of diameter D
- Does the conductance depend on
 - average diameter (D)
 - average area (D^2)
 - or more generally D^n (Sampson flow or Poiseuille flow)
- In this case, transformation may be an appropriate alternative to log transform





Outliers

Data points that we don't think are right

What is an outlier?

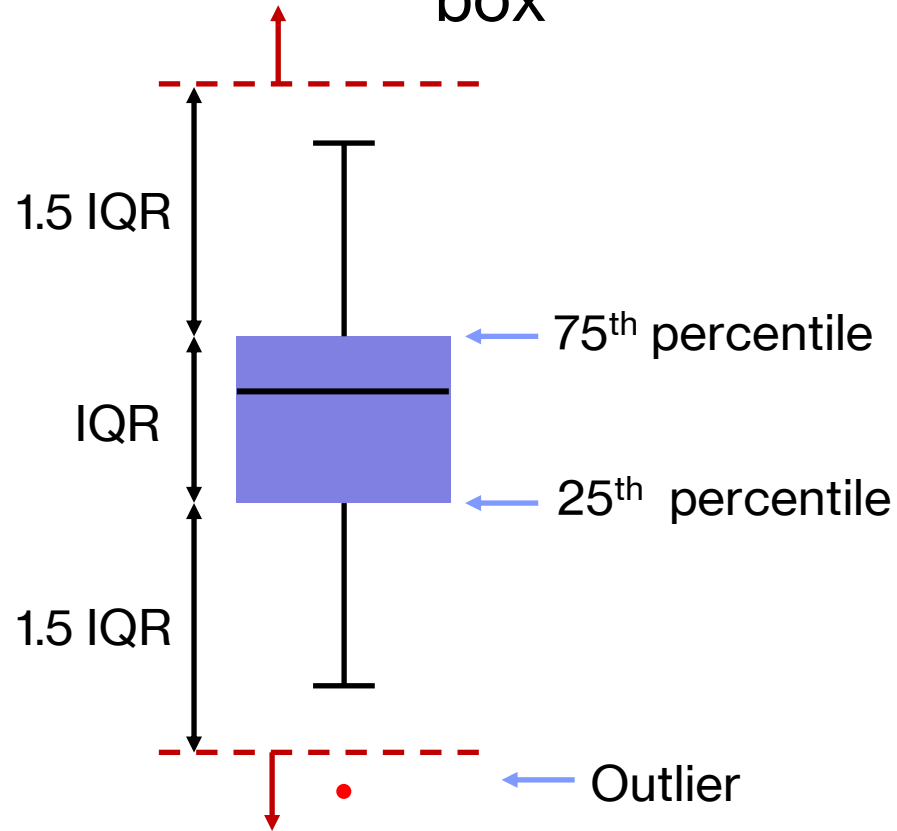
- Before investigating the distribution that the sample came from, it is important to omit any outliers
- Outliers can be caused by one of many things
 - Error in acquisition/measurement (look at your raw data)
 - Error in processing/analysis (check your working)
 - Abnormality in a response (common in biology)
 - Random sampling!
- The solution?
 - Reject them from the data set
- The problem
 - How to identify outliers fairly – objectively, rather than subjectively
- Some methods to identify outliers
 - Some number of SD from mean (z-score) – for normally distributed data
 - Some number of median absolute deviations from median
 - Some distance from IQR
- What isn't an outlier
 - A data point that you don't like!
 - A data point that makes $p > 0.05$ when you want it to be less! (*p-hacking*)



Identifying Outliers

Boxplot/Percentiles

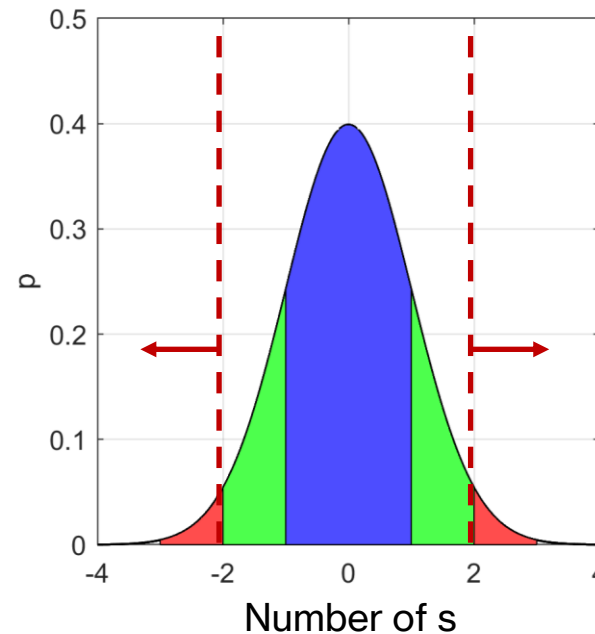
1.5 IQR away from the box



Approx Z-score

2 STD away from the mean

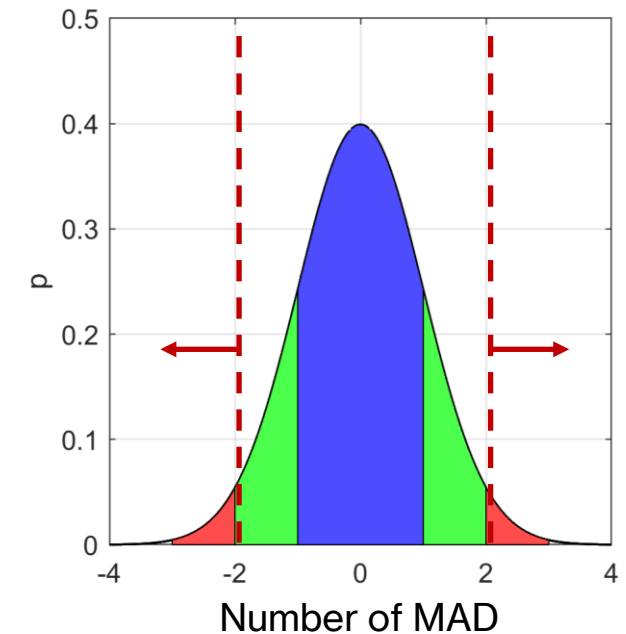
$$Z = \frac{x - \bar{x}}{s}$$



M(?) -score

2 MAD away from the median

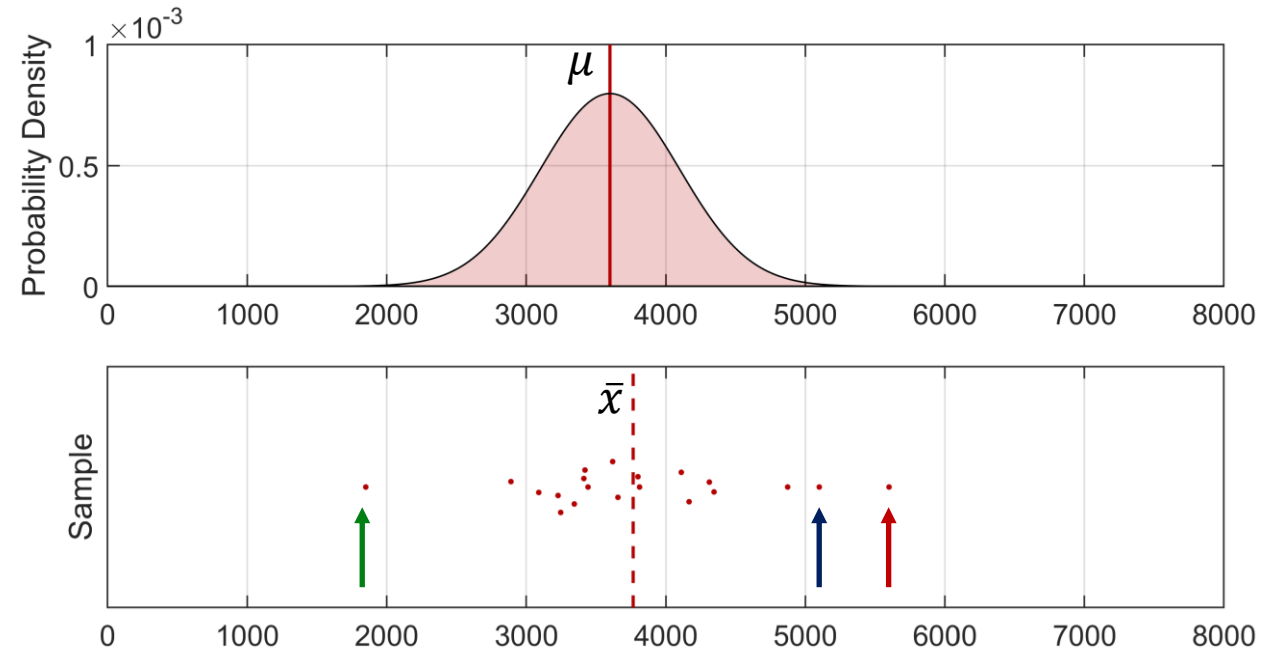
$$M = \frac{x - \tilde{x}}{MAD}$$



This distribution may not be normal

Simulating Outliers

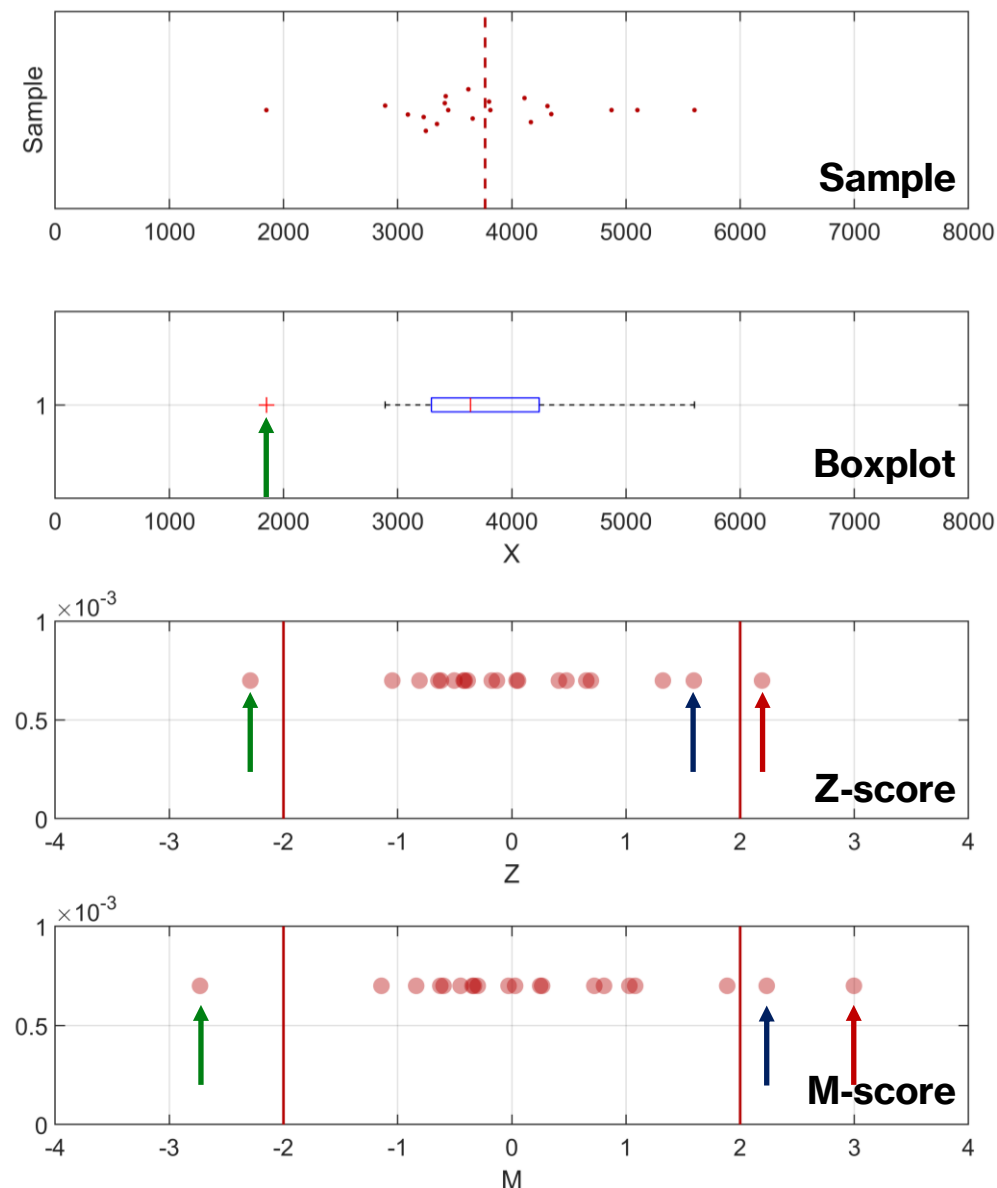
```
x=0:10000;  
mu2=3600;      % population mean  
sigma2=500;    % population sd  
n=20;  
% simulate the data  
X2=mu2+sigma2*randn(n-3,1);  
%forced outlier 1  
X2(n-2)=mu2-3.5*sigma2;  
% forced outlier 2  
X2(n-1)=mu2+3*sigma2;  
% forced outlier 3  
X2(n)=mu2+4*sigma2;
```



- Make a data point 3.5σ below μ
- Make a data point 3.0σ above μ
- Make a data point 4.0σ above μ

Outliers (2 above, 1 below)

- IQR approach identified 1/3
- Z-score approach identified 2/3
- M-score approach identified 3/3
- Caveat: it comes out differently for each simulation
- I recommended median absolute deviations from the median (M-score is made up terminology!)
- See [Leys et al. 2013](#) for more info
- Having removed outliers, we then check if our data fit our underlying assumptions...



Lecture 7: t-tests and assumptions

What we covered

- Degrees of freedom
- Confidence intervals with t-distribution
- T-tests
 - One sample t-test
 - Independent two-sample t-test
 - Dependent two-sample t-test
- Assumptions
 - Normality
 - Homoscedasticity
- Transformation
- Lognormal distribution
- Outliers

Next Up

Lecture 8: Non-parametric tests and regression